

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400590
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Seo; Satoshi et al.

Light-emitting element, light-emitting device, electronic device, and lighting device

Abstract

Emission efficiency of a light-emitting element is improved. The light-emitting element has a pair of electrodes and an EL layer between the pair of electrodes. The EL layer includes a first light-emitting layer and a second light-emitting layer. The first light-emitting layer includes a fluorescent material and a host material. The second light-emitting layer includes a phosphorescent material, a first organic compound, and a second organic compound. An emission spectrum of the second light-emitting layer has a peak in a yellow wavelength region. The first organic compound and the second organic compound form an exciplex.

Inventors: Seo; Satoshi (Kanagawa, JP), Ishisone; Takahiro (Kanagawa, JP), Ohsawa; Nobuharu (Kanagawa, JP), Nonaka; Yusuke (Kanagawa, JP), Sasaki; Toshiki (Kanagawa, JP)

Applicant: SEMICONDUCTOR ENERGY LABORATORY CO., LTD. (Atsugi, JP)

Family ID: 1000008778571

Assignee: Semiconductor Energy Laboratory Co., Ltd. (Atsugi, JP)

Appl. No.: 18/513763

Filed: November 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240164125 A1	May. 16, 2024

Related U.S. Application Data

continuation parent-doc US 17834973 20220608 US 11832465 child-doc US 18513763
continuation parent-doc US 16898514 20200611 US 11387422 20220712 child-doc US 17834973

continuation parent-doc US 16217231 20181212 US 10686152 20200616 child-doc US 16898514
continuation parent-doc US 15954681 20180417 US 10468619 20191105 child-doc US 16217231
continuation parent-doc US 15397790 20170104 US 9978971 20180522 child-doc US 15954681
continuation parent-doc US 14725026 20150529 US 9548468 20170117 child-doc US 15397790

Publication Classification

Int. Cl.: **H10K50/11** (20230101); **G09G3/3225** (20160101); **H10K50/12** (20230101);
H10K50/13 (20230101); **H10K50/15** (20230101); **H10K50/16** (20230101); **H10K50/81**
(20230101); **H10K50/82** (20230101); **H10K50/84** (20230101); **H10K59/00** (20230101);
H10K59/121 (20230101); **H10K59/131** (20230101); H10K59/38 (20230101);
H10K101/00 (20230101); H10K101/10 (20230101); H10K101/30 (20230101);
H10K101/40 (20230101)

U.S. Cl.:

CPC **H10K50/11** (20230201); **G09G3/3225** (20130101); **H10K50/12** (20230201);
H10K50/13 (20230201); **H10K50/131** (20230201); **H10K50/15** (20230201);
H10K50/16 (20230201); **H10K50/81** (20230201); **H10K50/82** (20230201); **H10K50/84**
(20230201); **H10K59/00** (20230201); **H10K59/1213** (20230201); **H10K59/131**
(20230201); G09G2300/0426 (20130101); H10K59/38 (20230201); H10K2101/10
(20230201); H10K2101/27 (20230201); H10K2101/30 (20230201); H10K2101/40
(20230201)

Field of Classification Search

CPC: H01L (51/5016); H01L (51/5004); H01L (51/5024); H01L (51/504); H01L (51/5044);
H01L (51/5056); H01L (51/5072); H01L (51/5206); H01L (51/5221); H01L (51/5237);
H01L (2251/5376); H01L (51/50-56); H01L (27/32-3297); H10K (50/131); H10K
(59/1213); H10K (59/131); H10K (50/828); H10K (50/00-88); H10K (59/00-95); H10K
(71/00-861); H10K (85/00-791); H10K (2101/00-90); H10K (2102/00-361); H10K
(85/341-348); G09G (2300/0452)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6238424	12/2000	Thiberg	N/A	N/A
6995509	12/2005	Yamazaki et al.	N/A	N/A
7009338	12/2005	D'Andrade et al.	N/A	N/A
7175922	12/2006	Jarikov et al.	N/A	N/A
7183010	12/2006	Jarikov	N/A	N/A
7285907	12/2006	D'Andrade et al.	N/A	N/A
7332857	12/2007	Seo et al.	N/A	N/A
7572522	12/2008	Seo et al.	N/A	N/A
7597967	12/2008	Kondakova et al.	N/A	N/A

7943925	12/2010	Yamazaki	N/A	N/A
7993760	12/2010	Komori et al.	N/A	N/A
8021574	12/2010	Kawamura et al.	N/A	N/A
8025815	12/2010	Kawamura et al.	N/A	N/A
8029697	12/2010	Kawamura et al.	N/A	N/A
8034256	12/2010	Kawamura et al.	N/A	N/A
8034465	12/2010	Liao et al.	N/A	N/A
8101857	12/2011	Kido et al.	N/A	N/A
8120020	12/2011	Kho et al.	N/A	N/A
8154195	12/2011	Nishimura et al.	N/A	N/A
8211552	12/2011	Nishimura et al.	N/A	N/A
8221907	12/2011	Kawamura et al.	N/A	N/A
8247086	12/2011	Inoue et al.	N/A	N/A
8274214	12/2011	Ikeda et al.	N/A	N/A
8294142	12/2011	Nishimura et al.	N/A	N/A
8330350	12/2011	Nishimura et al.	N/A	N/A
8436343	12/2012	Nishimura et al.	N/A	N/A
8568903	12/2012	Kawamura et al.	N/A	N/A
8587192	12/2012	Nishimura et al.	N/A	N/A
8653553	12/2013	Yamazaki et al.	N/A	N/A
8710493	12/2013	Nishimura et al.	N/A	N/A
8723025	12/2013	Kido et al.	N/A	N/A
8779655	12/2013	Nishimura et al.	N/A	N/A
8829517	12/2013	Hara et al.	N/A	N/A
8853680	12/2013	Yamazaki et al.	N/A	N/A
8921548	12/2013	Inoue et al.	N/A	N/A
8963127	12/2014	Pieh et al.	N/A	N/A
8981355	12/2014	Seo	N/A	N/A
8981393	12/2014	Seo et al.	N/A	N/A
8993129	12/2014	Endo et al.	N/A	N/A
8994263	12/2014	Shitagaki et al.	N/A	N/A
9054317	12/2014	Monkman et al.	N/A	N/A
9059421	12/2014	Seo et al.	N/A	N/A
9059430	12/2014	Seo et al.	N/A	N/A
9076976	12/2014	Seo et al.	N/A	N/A
9082995	12/2014	Nishimura et al.	N/A	N/A
9090532	12/2014	Kaiser et al.	N/A	N/A
9099617	12/2014	Yamazaki et al.	N/A	N/A
9142710	12/2014	Seo et al.	N/A	N/A
9153790	12/2014	Kuma et al.	N/A	N/A
9159942	12/2014	Seo et al.	N/A	N/A
9159957	12/2014	Pieh et al.	N/A	N/A
9175213	12/2014	Seo et al.	N/A	N/A
9184398	12/2014	Inoue et al.	N/A	N/A

9209415	12/2014	Shitagaki et al.	N/A	N/A
9219243	12/2014	Matsubara et al.	N/A	N/A
9263693	12/2015	Seo et al.	N/A	N/A
9263695	12/2015	Seo et al.	N/A	N/A
9299944	12/2015	Seo et al.	N/A	N/A
9356250	12/2015	Ohsawa et al.	N/A	N/A
9419237	12/2015	Shitagaki et al.	N/A	N/A
9502666	12/2015	Nakayama et al.	N/A	N/A
9525149	12/2015	Seo et al.	N/A	N/A
9559324	12/2016	Seo et al.	N/A	N/A
9590208	12/2016	Seo et al.	N/A	N/A
9604928	12/2016	Shitagaki et al.	N/A	N/A
9614170	12/2016	Ogiwara et al.	N/A	N/A
9627640	12/2016	Matsubara et al.	N/A	N/A
9634279	12/2016	Seo et al.	N/A	N/A
9660211	12/2016	Seo et al.	N/A	N/A
9680120	12/2016	Seo et al.	N/A	N/A
9786860	12/2016	Yamazaki et al.	N/A	N/A
9899620	12/2017	Ogiwara et al.	N/A	N/A
9929350	12/2017	Yamazaki et al.	N/A	N/A
9985223	12/2017	Inoue et al.	N/A	N/A
10062867	12/2017	Seo et al.	N/A	N/A
10121984	12/2017	Seo et al.	N/A	N/A
10164206	12/2017	Seo et al.	N/A	N/A
10181571	12/2018	Matsubara et al.	N/A	N/A
10361389	12/2018	Seo et al.	N/A	N/A
10361390	12/2018	Yamazaki et al.	N/A	N/A
10367160	12/2018	Seo et al.	N/A	N/A
10505135	12/2018	Seo et al.	N/A	N/A
10600972	12/2019	Seo et al.	N/A	N/A
10797257	12/2019	Seo et al.	N/A	N/A
10978661	12/2020	Seo et al.	N/A	N/A
11183644	12/2020	Seo et al.	N/A	N/A
2003/0175553	12/2002	Thompson et al.	N/A	N/A
2005/0048310	12/2004	Cocchi et al.	N/A	N/A
2005/0221116	12/2004	Cocchi et al.	N/A	N/A
2006/0134464	12/2005	Nariyuki	N/A	N/A
2006/0228577	12/2005	Nagara	N/A	N/A
2006/0232194	12/2005	Tung et al.	N/A	N/A
2007/0085106	12/2006	Kawakami et al.	N/A	N/A
2007/0090756	12/2006	Okada et al.	N/A	N/A
2007/0138947	12/2006	Popovic et al.	N/A	N/A
2007/0244320	12/2006	Inoue et al.	N/A	N/A
2008/0160345	12/2007	Inoue et al.	N/A	N/A
2008/0286604	12/2007	Inoue et al.	N/A	N/A
2008/0286610	12/2007	Deaton et al.	N/A	N/A
2009/0045731	12/2008	Nishimura et al.	N/A	N/A
2009/0179196	12/2008	Adachi et al.	N/A	N/A
2009/0302313	12/2008	Choi et al.	N/A	N/A

2010/0051968	12/2009	Seo et al.	N/A	N/A
2010/0052527	12/2009	Ikeda et al.	N/A	N/A
2010/0090209	12/2009	Ikari et al.	N/A	N/A
2010/0090593	12/2009	Mori et al.	N/A	N/A
2010/0145044	12/2009	Inoue et al.	N/A	N/A
2010/0171109	12/2009	Nishimura et al.	N/A	N/A
2010/0331585	12/2009	Kawamura et al.	N/A	N/A
2011/0001146	12/2010	Yamazaki et al.	N/A	N/A
2011/0108821	12/2010	Kaiser et al.	N/A	N/A
2011/0114889	12/2010	Buesing et al.	N/A	N/A
2011/0210316	12/2010	Kadoma et al.	N/A	N/A
2011/0215714	12/2010	Seo et al.	N/A	N/A
2012/0056170	12/2011	Pan et al.	N/A	N/A
2012/0097998	12/2011	Pieh et al.	N/A	N/A
2012/0098417	12/2011	Inoue et al.	N/A	N/A
2012/0126205	12/2011	Kawamura et al.	N/A	N/A
2012/0126208	12/2011	Kawamura et al.	N/A	N/A
2012/0126209	12/2011	Kawamura et al.	N/A	N/A
2012/0205632	12/2011	Shitagaki et al.	N/A	N/A
2012/0205687	12/2011	Yamazaki et al.	N/A	N/A
2012/0206035	12/2011	Shitagaki et al.	N/A	N/A
2012/0217486	12/2011	Takemura et al.	N/A	N/A
2012/0217487	12/2011	Yamazaki et al.	N/A	N/A
2012/0235127	12/2011	Takasu et al.	N/A	N/A
2012/0242219	12/2011	Seo et al.	N/A	N/A
2012/0248421	12/2011	Yamazaki et al.	N/A	N/A
2012/0256535	12/2011	Seo et al.	N/A	N/A
2012/0267618	12/2011	Monkman et al.	N/A	N/A
2012/0274201	12/2011	Seo et al.	N/A	N/A
2012/0277427	12/2011	Inoue et al.	N/A	N/A
2012/0305896	12/2011	Inoue	257/E51.026	H01L 51/009
2013/0009543	12/2012	Kadoma et al.	N/A	N/A
2013/0020561	12/2012	Suzuki et al.	N/A	N/A
2013/0048964	12/2012	Takeda et al.	N/A	N/A
2013/0049579	12/2012	Kaiser et al.	N/A	N/A
2013/0069090	12/2012	Sawabe et al.	N/A	N/A
2013/0088144	12/2012	Inoue et al.	N/A	N/A
2013/0207088	12/2012	Seo	N/A	N/A
2013/0240850	12/2012	Forrest et al.	N/A	N/A
2013/0240851	12/2012	Seo et al.	N/A	N/A
2013/0240933	12/2012	Yamazaki et al.	N/A	N/A
2013/0270531	12/2012	Seo et al.	N/A	N/A
2013/0277654	12/2012	Seo et al.	N/A	N/A
2013/0277655	12/2012	Seo et al.	N/A	N/A
2013/0277656	12/2012	Seo et al.	N/A	N/A

2013/0292656	12/2012	Seo et al.	N/A	N/A
2013/0292664	12/2012	Nishimura et al.	N/A	N/A
2013/0306945	12/2012	Seo	N/A	N/A
2014/0034926	12/2013	Matsubara et al.	N/A	N/A
2014/0034927	12/2013	Seo et al.	N/A	N/A
2014/0034929	12/2013	Hamada et al.	N/A	N/A
2014/0034930	12/2013	Seo et al.	N/A	N/A
2014/0034931	12/2013	Inoue et al.	N/A	N/A
2014/0034932	12/2013	Seo et al.	N/A	N/A
2014/0042469	12/2013	Seo et al.	N/A	N/A
2014/0061604	12/2013	Seo et al.	N/A	N/A
2014/0084274	12/2013	Yamazaki et al.	N/A	N/A
2014/0094246	12/2013	Pececnik	N/A	N/A
2014/0167027	12/2013	Yersin et al.	N/A	N/A
2014/0183499	12/2013	Kim	438/35	H01L 51/504
2014/0246071	12/2013	Kido et al.	N/A	N/A
2014/0252332	12/2013	Carroll	N/A	N/A
2014/0284578	12/2013	Takeda et al.	N/A	N/A
2014/0291646	12/2013	Shin	544/356	C07D 241/44
2014/0319492	12/2013	Seo et al.	N/A	N/A
2014/0361281	12/2013	Carroll	N/A	N/A
2015/0001502	12/2014	Seo et al.	N/A	N/A
2015/0028313	12/2014	Pillow et al.	N/A	N/A
2015/0053958	12/2014	Ishisone et al.	N/A	N/A
2015/0069352	12/2014	Kim et al.	N/A	N/A
2015/0155510	12/2014	Kawata et al.	N/A	N/A
2015/0155511	12/2014	Ohsawa et al.	N/A	N/A
2015/0171347	12/2014	Suzuki et al.	N/A	N/A
2015/0280158	12/2014	Ogiwara et al.	N/A	N/A
2015/0280163	12/2014	Seo et al.	N/A	N/A
2015/0333283	12/2014	Ishisone et al.	N/A	N/A
2016/0093823	12/2015	Seo et al.	N/A	N/A
2016/0268516	12/2015	Tanaka	N/A	C07D 471/06
2016/0343952	12/2015	Shitagaki et al.	N/A	N/A
2017/0025615	12/2016	Seo et al.	N/A	N/A
2017/0125705	12/2016	Nakayama et al.	N/A	N/A
2018/0212154	12/2017	Yamazaki et al.	N/A	N/A
2019/0140204	12/2018	Seo et al.	N/A	N/A
2019/0341568	12/2018	Yamazaki et al.	N/A	N/A
2020/0020870	12/2019	Seo et al.	N/A	N/A
2021/0013442	12/2020	Seo et al.	N/A	N/A
2021/0119150	12/2020	Seo et al.	N/A	N/A
2021/0226146	12/2020	Seo et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	---------------------	---------	-----

001953235	12/2006	CN	N/A
001988205	12/2006	CN	N/A
101771071	12/2009	CN	N/A
102076640	12/2010	CN	N/A
102076641	12/2010	CN	N/A
102456847	12/2011	CN	N/A
102655222	12/2011	CN	N/A
102757782	12/2011	CN	N/A
102884649	12/2012	CN	N/A
103052624	12/2012	CN	N/A
103204796	12/2012	CN	N/A
103443950	12/2012	CN	N/A
104170111	12/2013	CN	N/A
104247075	12/2013	CN	N/A
102008035413	12/2009	DE	N/A
102010020044	12/2010	DE	N/A
112012003073	12/2013	DE	N/A
112012005831	12/2014	DE	N/A
1202608	12/2001	EP	N/A
1777758	12/2006	EP	N/A
2166585	12/2009	EP	N/A
2166586	12/2009	EP	N/A
2166587	12/2009	EP	N/A
2166588	12/2009	EP	N/A
2166589	12/2009	EP	N/A
2166590	12/2009	EP	N/A
2166591	12/2009	EP	N/A
2204862	12/2009	EP	N/A
2571073	12/2012	EP	N/A
2004-522276	12/2003	JP	N/A
2004-241374	12/2003	JP	N/A
2006-024791	12/2005	JP	N/A
2006-024830	12/2005	JP	N/A
2006-120689	12/2005	JP	N/A
2007-103028	12/2006	JP	N/A
2007-173827	12/2006	JP	N/A
2008-288344	12/2007	JP	N/A
2010-157722	12/2009	JP	N/A
2011-529455	12/2010	JP	N/A
2011-529614	12/2010	JP	N/A
2012-004526	12/2011	JP	N/A
2012-149030	12/2011	JP	N/A
2012-186461	12/2011	JP	N/A
2012-195573	12/2011	JP	N/A
2012-238854	12/2011	JP	N/A
2013-047283	12/2012	JP	N/A
2013-048221	12/2012	JP	N/A
2013-065428	12/2012	JP	N/A
2013-093570	12/2012	JP	N/A
2013-100341	12/2012	JP	N/A

2013-526773	12/2012	JP	N/A
2013-219024	12/2012	JP	N/A
2013-219347	12/2012	JP	N/A
2013-239431	12/2012	JP	N/A
2013-239704	12/2012	JP	N/A
2014-006519	12/2013	JP	N/A
2014-043437	12/2013	JP	N/A
2014-044941	12/2013	JP	N/A
2014-044942	12/2013	JP	N/A
2014-056814	12/2013	JP	N/A
2014-075249	12/2013	JP	N/A
2014-078699	12/2013	JP	N/A
2014-143437	12/2013	JP	N/A
2016-072632	12/2015	JP	N/A
2017-028283	12/2016	JP	N/A
2017-120867	12/2016	JP	N/A
2021-040150	12/2020	JP	N/A
2007-0067606	12/2006	KR	N/A
2010-0012781	12/2009	KR	N/A
2010-0012808	12/2009	KR	N/A
2010-0076813	12/2009	KR	N/A
2012-0043606	12/2011	KR	N/A
2012-0100751	12/2011	KR	N/A
2013-0037646	12/2012	KR	N/A
2013-0107206	12/2012	KR	N/A
2014-0015259	12/2013	KR	N/A
2014-0018146	12/2013	KR	N/A
2014-0026577	12/2013	KR	N/A
2014-0135975	12/2013	KR	N/A
2014-0136027	12/2013	KR	N/A
2015-0002674	12/2014	KR	N/A
2015-0038501	12/2014	KR	N/A
2015-0060976	12/2014	KR	N/A
2018-0031696	12/2017	KR	N/A
2018-0040130	12/2017	KR	N/A
201125165	12/2010	TW	N/A
201213497	12/2011	TW	N/A
201242132	12/2011	TW	N/A
201307285	12/2012	TW	N/A
201307528	12/2012	TW	N/A
201313682	12/2012	TW	N/A
201314985	12/2012	TW	N/A
201347265	12/2012	TW	N/A
201349614	12/2012	TW	N/A
201349619	12/2012	TW	N/A
201401602	12/2013	TW	N/A
201407852	12/2013	TW	N/A
201407854	12/2013	TW	N/A
201415688	12/2013	TW	N/A
201711244	12/2016	TW	N/A

WO-2000/070655	12/1999	WO	N/A
WO-2009/008344	12/2008	WO	N/A
WO-2010/012328	12/2009	WO	N/A
WO-2010/012330	12/2009	WO	N/A
WO-2011/027653	12/2010	WO	N/A
WO-2011/141109	12/2010	WO	N/A
WO-2012/011258	12/2011	WO	N/A
WO-2012/053627	12/2011	WO	N/A
WO-2012/099241	12/2011	WO	N/A
WO-2012/127990	12/2011	WO	N/A
WO-2013/015144	12/2012	WO	N/A
WO-2013/137088	12/2012	WO	N/A
WO-2013/137089	12/2012	WO	N/A
WO-2013/157506	12/2012	WO	N/A
WO-2013/157559	12/2012	WO	N/A
WO-2014/021443	12/2013	WO	N/A
WO-2014/046221	12/2013	WO	N/A
WO-2014/054452	12/2013	WO	N/A
WO-2014/185434	12/2013	WO	N/A
WO-2016/051309	12/2015	WO	N/A
WO-2017/013526	12/2016	WO	N/A

OTHER PUBLICATIONS

Baldo.M et al., “Highly Efficient Phosphorescent Emission From Organic Electroluminescent Devices”, Nature, Sep. 10, 1998, vol. 395, pp. 151-154. cited by applicant

Baldo.M et al., “Very High-Efficiency Green Organic Light-Emitting Devices Based on Electrophosphorescence”, Appl. Phys. Lett. (Applied Physics Letters) , Jul. 5, 1999, vol. 75, No. 1, pp. 4-6. cited by applicant

Gu.G et al., “Transparent Organic Light Emitting Devices”, Appl. Phys. Lett. (Applied Physics Letters) , May 6, 1996, vol. 68, No. 19, pp. 2606-2608. cited by applicant

Choong.V et al., “Organic Light-Emitting Diodes With a Bipolar Transport Layer”, Appl. Phys. Lett. (Applied Physics Letters) , Jul. 12, 1999, vol. 75, No. 2, pp. 172-174. cited by applicant

Adachi.C et al., “Nearly 100% Internal Phosphorescence Efficiency in an Organic Light Emitting Device”, J. Appl. Phys. (Journal of Applied Physics) , Nov. 15, 2001, vol. 90, No. 10, pp. 5048-5051. cited by applicant

Baldo.M et al., “Prospects for Electrically Pumped Organic Lasers”, Phys. Rev. B (Physical Review. B), Jul. 19, 2002, vol. 66, pp. 035321-1-035321-16. cited by applicant

Markham.J et al., “High-efficiency green phosphorescence from spin-coated single-layer dendrimer light-emitting diodes”, Appl. Phys. Lett. (Applied Physics Letters) , Apr. 15, 2002, vol. 80, No. 15, pp. 2645-2647. cited by applicant

Itano.K et al., “Exciplex formation at the organic solid-state interface: Yellow emission in organic light-emitting diodes using green-fluorescent tris(8-quinolinolato)aluminum and hole-transporting molecular materials with low ionization potentials”, Appl. Phys. Lett. (Applied Physics Letters) , Feb. 9, 1998, vol. 72, No. 6, pp. 636-638. cited by applicant

Kondakova.M et al., “High-efficiency, low-voltage phosphorescent organic light-emitting diode devices with mixed host”, J. Appl. Phys. (Journal of Applied Physics) , Nov. 4, 2008, vol. 104, pp. 094501-1-094501-17. cited by applicant

Seo.J et al., “Efficient blue-green organic light-emitting diodes based on heteroleptic tris-cyclometalated iridium(III) complexes”, Thin Solid Films, Sep. 25, 2008, vol. 517, No. 5, pp. 1807-1810. cited by applicant

Goushi.K et al., "Efficient organic light-emitting diodes through up-conversion from triplet to singlet excited states of exciplexes", Appl. Phys. Lett. (Applied Physics Letters) , Jul. 12, 2012, vol. 101, No. 2, pp. 023306-1-023306-4. cited by applicant

Fujita.M et al., "Reduction of operating voltage in organic light-emitting diode by corrugated photonic crystal structure", Appl. Phys. Lett. (Applied Physics Letters) , Dec. 6, 2004, vol. 85, No. 23, pp. 5769-5771. cited by applicant

Hirakata.Y et al., "Development of High-Definition organic light emitting display", Optronics, Dec. 10, 2013, No. 384, pp. 33-39. cited by applicant

International Search Report (Application No. PCT/IB2015/053577) Dated Sep. 1, 2015. cited by applicant

Written Opinion (Application No. PCT/IB2015/053577) Dated Sep. 1, 2015. cited by applicant

Yersin.H et al., Highly Efficient OLEDs with Phosphorescent Materials, 2008, pp. 1-97,283-309, Wiley-VCH Verlag GmbH & Co cited by applicant

Tokito.S et al., "Improvement in performance by doping", Organic EL Display, Aug. 20, 2004, pp. 67-99, Ohmsha. cited by applicant

Jeon.W et al., "Ideal host and guest system in phosphorescent OLEDs", Organic Electronics, 2009, vol. 10, pp. 240-246, Elsevier. cited by applicant

Su.S et al., "RGB Phosphorescent Organic Light-Emitting Diodes by Using Host Materials with Heterocyclic Cores:Effect of Nitrogen Atom Orientations", Chem. Mater. (Chemistry of Materials), 2011, vol. 23, No. 2, pp. 274-284. cited by applicant

Rausch.A et al., "Matrix Effects on the Triplet State of the OLED Emitter Ir(4,6-dFppy)2(pic) (Flrpic): Investigations by High-Resolution Optical Spectroscopy", Inorg. Chem. (Inorganic Chemistry), 2009, vol. 48, No. 5, pp. 1928-1937. cited by applicant

Gong.X et al., "Phosphorescence from iridium complexes doped into polymer blends", J. Appl. Phys. (Journal of Applied Physics) , Feb. 1, 2004, vol. 95, No. 3, pp. 948-953. cited by applicant

Zhao.Q et al., "Synthesis and Photophysical, Electrochemical, and Electrophosphorescent Properties of a Series of Iridium(III) Complexes Based on Quinoline Derivatives and Different β -Diketonate Ligands", Organometallics, Jun. 14, 2006, vol. 25, No. 15, pp. 3631-3638. cited by applicant

Hino.Y et al., "Red Phosphorescent Organic Light-Emitting Diodes Using Mixture System of Small-Molecule and Polymer Host", Jpn. J. Appl. Phys. (Japanese Journal of Applied Physics) , Apr. 21, 2005, vol. 44, No. 4B, pp. 2790-2794, The Japan Society of Applied Physics. cited by applicant

Tsuboyama.A et al., "Homoleptic Cyclometalated Iridium Complexes with Highly Efficient Red Phosphorescence and Application to Organic Light-Emitting Diode", J. Am. Chem. Soc. (Journal of the American Chemical Society), 2003, vol. 125, No. 42, pp. 12971-12979. cited by applicant

Chen.F et al., "Triplet Exciton Confinement in Phosphorescent Polymer Light-Emitting Diodes", Appl. Phys. Lett. (Applied Physics Letters) , Feb. 17, 2003, vol. 82, No. 7, pp. 1006-1008. cited by applicant

Lee.J et al., "Stabilizing the efficiency of phosphorescent organic light-emitting diodes", SPIE Newsroom, Apr. 21, 2008, pp. 1-3. cited by applicant

Tokito.S et al., "Confinement of Triplet Energy on Phosphorescent Molecules for Highly-Efficient Organic Blue-Light-Emitting Devices", Appl. Phys. Lett. (Applied Physics Letters) , Jul. 21, 2003, vol. 83, No. 3, pp. 569-571. cited by applicant

Endo.A et al., "Efficient Up-Conversion of Triplet Excitons Into a Singlet State and Its Application for Organic Light Emitting Diodes", Appl. Phys. Lett. (Applied Physics Letters) , Feb. 24, 2011, vol. 98, No. 8, pp. 083302-1-083302-3. cited by applicant

Park.Y et al., "Efficient triplet harvesting by fluorescent molecules through exciplexes for high efficiency organic light-emitting diodes", Appl. Phys. Lett. (Applied Physics Letters) , Apr. 18, 2013, vol. 102, No. 15, pp. 153306-1-153306-5. cited by applicant

Primary Examiner: Ahmed; Shahed

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/834,973, filed Jun. 8, 2022, now allowed, which is a continuation of U.S. application Ser. No. 16/898,514, filed Jun. 11, 2020, now U.S. Pat. No. 11,387,422, which is a continuation of U.S. application Ser. No. 16/217,231, filed Dec. 12, 2018, now U.S. Pat. No. 10,686,152, which is a continuation of U.S. application Ser. No. 15/954,681, filed Apr. 17, 2018, now U.S. Pat. No. 10,468,619, which is a continuation of U.S. application Ser. No. 15/397,790, filed Jan. 4, 2017, now U.S. Pat. No. 9,978,971, which is a continuation of U.S. application Ser. No. 14/725,026, filed May 29, 2015, now U.S. Pat. No. 9,548,468, which claims the benefit of foreign priority applications filed in Japan as Serial No. 2014-112448 on May 30, 2014, and Serial No. 2014-241137 on Nov. 28, 2014, all of which are incorporated by reference.

TECHNICAL FIELD

(1) One embodiment of the present invention relates to a light-emitting element in which a light-emitting layer capable of emitting light by application of an electric field is provided between a pair of electrodes, and also relates to a light-emitting device, an electronic device, and a lighting device including the light-emitting element.

(2) Note that one embodiment of the present invention is not limited to the above technical field. The technical field of one embodiment of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. In addition, one embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter. Specifically, examples of the technical field of one embodiment of the present invention disclosed in this specification include a semiconductor device, a display device, a liquid crystal display device, a light-emitting device, a lighting device, a power storage device, a storage device, a method for driving any of them, and a method for manufacturing any of them.

BACKGROUND ART

(3) In recent years, research and development of a light-emitting element (organic EL element) which uses an organic compound and utilizes electroluminescence (EL) have been actively promoted. In the basic structure of such a light-emitting element, an organic compound layer containing a light-emitting substance (an EL layer) is provided between a pair of electrodes. By voltage application to this element, light emission from the light-emitting substance can be obtained.

(4) A light-emitting element in which an organic compound layer is between a pair of electrodes is referred to as an organic electroluminescence element, and a light-emitting device including the light-emitting element is referred to as an organic electroluminescence device. The organic electroluminescence device can be used in a display device, a lighting device, and the like (see Patent Document 1, for example).

REFERENCE

Patent Document

(5) [Patent Document 1] Japanese Published Patent Application No. 2012-186461

DISCLOSURE OF INVENTION

(6) An object of one embodiment of the present invention is to improve emission efficiency of a light-emitting element. Another object of one embodiment of the present invention is to provide a novel semiconductor device, a novel light-emitting element, or a novel light-emitting device. Note that the descriptions of these objects do not disturb the existence of other objects. In one embodiment of the present invention, there is no need to achieve all the objects. Other objects will be apparent from and can be derived from the description of the specification, the drawings, the claims, and the like.

(7) One embodiment of the present invention is a light-emitting element in which an EL layer is between a pair of electrodes. The EL layer includes a first light-emitting layer and a second light-emitting layer. The first light-emitting layer includes a fluorescent material and a host material. The second light-emitting layer includes a phosphorescent material, a first organic compound, and a second organic compound. An emission spectrum of the second light-emitting layer has a peak in a yellow wavelength region. The first organic compound and the second organic compound form an exciplex.

(8) In the above structure, it is preferred that the second light-emitting layer include one phosphorescent material.

(9) In any of the above structures, it is preferred that energy be transferred from the exciplex to the phosphorescent material.

(10) In any of the above structures, it is preferred that a singlet excited level of the host material be higher than that of the fluorescent material and a triplet excited level of the host material be lower than that of the fluorescent material.

(11) In any of the above structures, it is preferred that a triplet excited level of the host material be lower than those of the first organic compound and the second organic compound.

(12) In any of the above structures, it is preferred that the first light-emitting layer and the second light-emitting layer include a region where the first light-emitting layer and the second light-emitting layer are in contact with each other.

(13) In any of the above structures, it is preferred that the first light-emitting layer and the second light-emitting layer include a region where the first light-emitting layer and the second light-emitting layer are separated from each other, in which case a mixed layer of a hole-transport material and an electron-transport material is preferably provided between the first light-emitting layer and the second light-emitting layer.

(14) In any of the above structures, it is preferred that the second light-emitting layer be over the first light-emitting layer.

(15) Another embodiment of the present invention is a light-emitting device including the light-emitting element with any of the above structures, and a transistor or a substrate.

(16) The light-emitting device in this specification and the like includes an image display device that uses a light-emitting element. Furthermore, the light-emitting device may be included in a module in which a light-emitting element is provided with a connector such as a flexible printed circuit (FPC), a module in which a light-emitting element is provided with an anisotropic conductive film or a tape carrier package (TCP), a module in which a printed wiring board is provided at the end of the TCP, or a module in which an integrated circuit (IC) is directly mounted on a light-emitting element by a chip on glass (COG) method.

(17) Another embodiment of the present invention is an electronic device including a light-emitting device with the above structure and an external connection port, a keyboard, an operation button, a speaker, or a microphone. Another embodiment of the present invention is an electronic device including a module with the above structure and an external connection port, a keyboard, an operation button, a speaker, or a microphone. Another embodiment of the present invention is a lighting device including a light-emitting device with the above structure and a housing.

(18) According to one embodiment of the present invention, emission efficiency of a light-emitting element can be improved. According to one embodiment of the present invention, a novel

semiconductor device, a novel light-emitting element, or a novel light-emitting device can be provided.

(19) Note that the description of these effects does not disturb the existence of other effects. One embodiment of the present invention does not necessarily achieve all the effects listed above. Other effects will be apparent from and can be derived from the description of the specification, the drawings, the claims, and the like.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) In the accompanying drawings:
- (2) FIGS. 1A and 1B are schematic cross-sectional views illustrating light-emitting elements of one embodiment of the present invention;
- (3) FIGS. 2A and 2B show correlations of energy levels in light-emitting layers;
- (4) FIG. 3 shows a correlation of energy levels in light-emitting layers;
- (5) FIGS. 4A and 4B are schematic cross-sectional views illustrating light-emitting elements of one embodiment of the present invention;
- (6) FIG. 5 is a schematic cross-sectional view illustrating a light-emitting element of one embodiment of the present invention;
- (7) FIGS. 6A and 6B are a top view and a cross-sectional view, respectively, illustrating a light-emitting device of one embodiment of the present invention;
- (8) FIGS. 7A and 7B are cross-sectional views illustrating light-emitting devices of one embodiment of the present invention;
- (9) FIG. 8 is a cross-sectional view illustrating a light-emitting device of one embodiment of the present invention;
- (10) FIGS. 9A and 9B are cross-sectional views illustrating light-emitting devices of one embodiment of the present invention;
- (11) FIGS. 10A and 10B are cross-sectional views illustrating light-emitting devices of one embodiment of the present invention;
- (12) FIGS. 11A and 11B are a block diagram and a circuit diagram, respectively, illustrating a display device of one embodiment of the present invention;
- (13) FIG. 12 is a perspective view illustrating a display module;
- (14) FIGS. 13A to 13G each illustrate an electronic device;
- (15) FIG. 14 illustrates lighting devices;
- (16) FIGS. 15A and 15B are schematic cross-sectional views illustrating element structures of light-emitting elements of Examples 1 to 3;
- (17) FIGS. 16A and 16B show current density-luminance characteristics and voltage-luminance characteristics, respectively, of light-emitting elements of Example 1;
- (18) FIGS. 17A and 17B show luminance-power efficiency characteristics and luminance-current efficiency characteristics, respectively, of light-emitting elements of Example 1;
- (19) FIG. 18 shows emission spectra of light-emitting elements of Example 1;
- (20) FIGS. 19A and 19B show current density-luminance characteristics and voltage-luminance characteristics, respectively, of a light-emitting element of Example 2;
- (21) FIGS. 20A and 20B show luminance-power efficiency characteristics and luminance-current efficiency characteristics, respectively, of a light-emitting element of Example 2;
- (22) FIG. 21 shows an emission spectrum of a light-emitting element of Example 2;
- (23) FIGS. 22A and 22B show current density-luminance characteristics and voltage-luminance characteristics, respectively, of a light-emitting element of Example 3;
- (24) FIGS. 23A and 23B show luminance-power efficiency characteristics and luminance-current

- efficiency characteristics, respectively, of a light-emitting element of Example 3;
- (25) FIG. 24 shows an emission spectrum of a light-emitting element in Example 3;
- (26) FIG. 25 is a schematic cross-sectional view illustrating an element structure of a light-emitting element of Example 4;
- (27) FIGS. 26A and 26B show current density-luminance characteristics and voltage-luminance characteristics, respectively, of light-emitting elements of Example 4;
- (28) FIG. 27 shows emission spectra of light-emitting elements of Example 4;
- (29) FIG. 28 shows luminance-external quantum efficiency characteristics of a light-emitting element 11 of Example 5;
- (30) FIG. 29 shows an emission spectrum of a light-emitting element 11 of Example 5;
- (31) FIG. 30 shows reliability of a light-emitting element 11 of Example 5;
- (32) FIG. 31 shows luminance-external quantum efficiency characteristics of a light-emitting element 12 of Example 5;
- (33) FIG. 32 shows an emission spectrum of a light-emitting element 12 of Example 5; and
- (34) FIG. 33 shows reliability of a light-emitting element 12 of Example 5.

BEST MODE FOR CARRYING OUT THE INVENTION

(35) Embodiments of the present invention will be explained below with reference to the drawings. Note that one embodiment of the invention is not limited to the description given below, and various changes and modifications can be made without departing from the spirit and scope of the invention. Therefore, one embodiment of the present invention is not interpreted as being limited to the description of the embodiments described below.

(36) Note that the position, the size, the range, or the like of each structure illustrated in drawings and the like is not accurately represented in some cases for simplification. Therefore, the disclosed invention is not necessarily limited to the position, the size, the range, or the like disclosed in the drawings and the like.

(37) Note that the ordinal numbers such as “first” and “second” in this specification and the like are used for convenience and do not denote the order of steps or the stacking order of layers.

Therefore, for example, description can be made even when “first” is replaced with “second” or “third”, as appropriate. In addition, the ordinal numbers in this specification and the like are not necessarily the same as those which specify one embodiment of the present invention.

(38) In describing structures of the present invention with reference to the drawings, the same reference numerals are used in common for the same portions in different drawings in this specification and the like.

(39) Note that the terms “film” and “layer” can be interchanged with each other depending on the case or circumstances in this specification and the like. For example, the term “conductive layer” can be changed into the term “conductive film” in some cases. Also, the term “insulating film” can be changed into the term “insulating layer” in some cases.

(40) In this specification and the like, a fluorescent material refers to a material that emits light in the visible light region when the level of the lowest singlet excited state (S.sub.1 level) relaxes to the ground state. A phosphorescent material refers to a material that emits light in the visible light region at room temperature when the level of the lowest triplet excited state (T.sub.1 level) relaxes to the ground state. That is, a phosphorescent material refers to a material that can convert triplet excitation energy into visible light.

(41) In this specification and the like, blue light has at least one peak of emission spectrum in a blue wavelength region of greater than or equal to 420 nm and less than or equal to 480 nm, green light has at least one peak of emission spectrum in a green wavelength region of greater than or equal to 500 nm and less than 550 nm, yellow light has at least one peak of emission spectrum in a yellow wavelength region of greater than or equal to 550 nm and less than or equal to 590 nm, and red light has at least one peak of emission spectrum in a red wavelength region of greater than or equal to 600 nm and less than or equal to 740 nm.

Embodiment 1

(42) Light-emitting elements of one embodiment of the present invention are described with reference to FIGS. 1A and 1B. FIG. 1A is a schematic cross-sectional view of a light-emitting element **100** of one embodiment of the present invention, and FIG. 1B is a schematic cross-sectional view of a light-emitting element **140** of one embodiment of the present invention.

(43) In the light-emitting element **100** shown in FIG. 1A, an EL layer **130** is between a pair of electrodes (a first electrode **101** and a second electrode **102**). The EL layer **130** includes a first light-emitting layer **113** and a second light-emitting layer **114**. In the light-emitting element **100**, a hole-injection layer **111**, a hole-transport layer **112**, an electron-transport layer **115**, and an electron-injection layer **116** are illustrated as part of the EL layer **130**. However, this stacked-layer structure is an example, and the structure of the EL layer **130** in the light-emitting element of one embodiment of the present invention is not limited thereto. Note that, in the light-emitting element **100**, the first electrode **101** serves as an anode, and the second electrode **102** serves as a cathode.

(44) The first light-emitting layer **113** includes a fluorescent material and a host material. An emission spectrum of the first light-emitting layer **113** preferably has a peak in a blue wavelength region. The second light-emitting layer **114** includes a phosphorescent material, a first organic compound, and a second organic compound. An emission spectrum of the second light-emitting layer **114** preferably has a peak in a yellow wavelength region. The second light-emitting layer **114** preferably includes one phosphorescent material. The first organic compound and the second organic compound form an exciplex. One of the first organic compound and the second organic compound serves as a host material for the second light-emitting layer **114**, and the other of the first organic compound and the second organic compound serves as an assist material for the second light-emitting layer **114**. Note that the first organic compound serves as the host material and the second organic compound serves as the assist material in the following description.

(45) When the first light-emitting layer **113** and the second light-emitting layer **114** have the above structures, fluorescent light emission from the first light-emitting layer **113** (here, light emission with a peak in the blue wavelength region) and phosphorescent light emission from the second light-emitting layer **114** (here, light emission with a peak in the yellow wavelength region) can be efficiently obtained.

(46) A T.sub.1 level of the host material of the first light-emitting layer **113** is preferably lower than T.sub.1 levels of the first and second organic compounds of the second light-emitting layer **114**. In the first light-emitting layer **113**, an S.sub.1 level of the host material is preferably higher than an S.sub.1 level of the fluorescent material while the T.sub.1 level of the host material is lower than a T.sub.1 level of the fluorescent material.

(47) Although there is no limitation on the combination of the first organic compound and the second organic compound in the second light-emitting layer **114** as long as an exciplex can be formed, it is preferred that one organic compound be a material having a hole-transport property and the other organic compound be a material having an electron-transport property. In that case, a donor-acceptor excited state is formed easily, which allows an exciplex to be formed efficiently. In the case where the combination of the first organic compound and the second organic compound is a combination of the material having a hole-transport property and the material having an electron-transport property, the carrier balance can be easily controlled depending on the mixture ratio. Specifically, the ratio of the material having a hole-transport property to the material having an electron-transport property is preferably within a range of 1:9 to 9:1 (weight ratio). Since the carrier balance can be easily controlled in the light-emitting element **100** having the structure, a recombination region can also be easily adjusted.

(48) In the light-emitting element **100**, a carrier recombination region is preferably distributed to some extent. Therefore, it is preferred that the first light-emitting layer **113** or the second light-emitting layer **114** have an appropriate degree of carrier-trapping property. It is particularly preferred that the phosphorescent material in the second light-emitting layer **114** have an electron-

trapping property.

(49) Note that in the light-emitting element **100**, light emitted from the first light-emitting layer **113** preferably has a peak on the shorter wavelength side than light emitted from the second light-emitting layer **114**. The luminance of a light-emitting element using the phosphorescent material emitting light with a short wavelength tends to degrade quickly. In view of the above, fluorescence is used for light emission with a short wavelength, so that a light-emitting element with less degradation of luminance can be provided.

(50) Because the first light-emitting layer **113** and the second light-emitting layer **114** are stacked to be in contact with each other in the light-emitting element **100**, the number of layers for forming the EL layer **130** is small and productivity is high.

(51) Further, in the light-emitting element **100**, the first light-emitting layer **113** and the second light-emitting layer **114** are made to emit light with different emission wavelengths, so that the light-emitting element can be a multicolor light-emitting element. The emission spectrum of the light-emitting element **100** is formed by combining light having different emission peaks, and thus has at least two peaks.

(52) The light-emitting element **100** is suitable for obtaining white light emission. When the first light-emitting layer **113** and the second light-emitting layer **114** emit light of complementary colors, white light emission can be obtained.

(53) In addition, white light emission with a high color rendering property that is formed of three primary colors or four or more colors can be obtained by using a plurality of light-emitting substances emitting light with different wavelengths for the first light-emitting layer **113**. In that case, the first light-emitting layer **113** may be divided into layers and each of the divided layers may contain a different light-emitting substance from the others.

Light Emission Mechanism of Second Light-Emitting Layer

(54) FIG. 2A shows a correlation of energy levels between the first organic compound, the second organic compound, and the phosphorescent material of the second light-emitting layer **114**. The following explains what terms and signs in FIG. 2A represent: Host: the first organic compound; Assist: the second organic compound; Guest: the phosphorescent material; S.sub.PH: the level of the lowest singlet excited state of the host material (the first organic compound); T.sub.PH: the level of the lowest triplet excited state of the host material (the first organic compound); T.sub.PG: the level of the lowest triplet excited state of the guest material (the phosphorescent material); S.sub.E: the level of the lowest singlet excited state of the exciplex; and T.sub.E: the level of the lowest triplet excited state of the exciplex.

(55) In the light-emitting element **100** of one embodiment of the present invention, the first and second organic compounds of the second light-emitting layer **114** form the exciplex. The level of the lowest singlet excited state of the exciplex (S.sub.E) and the level of the lowest triplet excited state of the exciplex (T.sub.E) are adjacent to each other (see Route A in FIG. 2A).

(56) An exciplex is an excited state formed from two kinds of substances. In the case of photoexcitation, the exciplex is formed in such a manner that one molecule in an excited state takes in the other substance in a ground state. The two kinds of substances that have formed the exciplex return to a ground state by emitting light and serve as the original two kinds of substances. In the case of electrical excitation, the exciplex can be formed when a cationic molecule (hole) of one substance comes close to an anionic molecule (electron) of the other substance. That is, the exciplex can be formed without formation of excitation state of any molecule in the electrical excitation; thus, a driving voltage can be lowered. Both energies of S.sub.E and T.sub.E of the exciplex then move to the level of the lowest triplet excited state of the guest material (the phosphorescent material) to obtain light emission (see Route B in FIG. 2A).

(57) The above-described process of Route A and Route B is referred to as exciplex-triplet energy transfer (ExTET) in this specification and the like. As described, in the light-emitting element of one embodiment of the present invention, energy can be given from the exciplex to the

phosphorescent material (guest material).

(58) When one of the first and second organic compounds receiving a hole and the other of the first and second organic compounds receiving an electron come close to each other, the exciplex is formed at once. Alternatively, when one substance becomes in an excited state, the one immediately takes in the other substance to form the exciplex. Therefore, most excitons in the second light-emitting layer **114** exist as the exciplexes. A band gap of the exciplex is narrower than those of the first organic compound and the second organic compound; therefore, the driving voltage can be lowered when the exciplex is formed by recombination of a hole and an electron.

Light Emission Mechanism of First Light-Emitting Layer

(59) In the first light-emitting layer **113**, recombination of carriers forms an excited state. Note that the first light-emitting layer **113** includes the host material and the fluorescent material. Because the amount of the host material is large as compared to the fluorescent material, the excited states are formed mostly as the excited states of the host material. The ratio of singlet excited states to triplet excited states caused by carrier recombination (hereinafter referred to as exciton generation probability) is approximately 1:3.

(60) First, a case where the T.sub.1 level of the host material is higher than the T.sub.1 level of the guest material is described below.

(61) Energy is transferred from the host material in the triplet excited state to the guest material (triplet energy transfer). However, the triplet excited state of the guest material does not offer emission of light in a visible light region because the guest material is the fluorescent material. Thus, the triplet excited state of the host material cannot be used for light emission. Therefore, when the T.sub.1 level of the host material is higher than the T.sub.1 level of the guest material, only approximately 25% of injected carriers can be used for light emission at most.

(62) FIG. 2B shows a correlation of energy levels between the host material and the fluorescent material of the first light-emitting layer **113**. The following explains what terms and signs in FIG. 2B represent: Host: the host material; Guest: the fluorescent material; S.sub.FH: the level of the lowest singlet excited state of the host material; T.sub.FH: the level of the lowest triplet excited state of the host material; S.sub.FG: the level of the lowest singlet excited state of the guest material (the fluorescent material); and T.sub.FG: the level of the lowest triplet excited state of the guest material (the fluorescent material).

(63) As shown in FIG. 2B, the T.sub.1 level of the guest material (T.sub.FG in FIG. 2B) is higher than the T.sub.1 level of the guest material (T.sub.FE in FIG. 2B).

(64) In addition, as shown in FIG. 2B, triplet excitons collide with each other by triplet-triplet annihilation (TTA), and part of energy of them is converted into the level of the lowest singlet excited state of the host material (S.sub.FH). Energy is transferred from the level of the lowest singlet excited state of the host material (S.sub.FH) to the level of the lowest singlet excited state of the guest material (the fluorescent material) (S.sub.FG) that is the level lower than S.sub.FH (see Route C in FIG. 2B); and thus the fluorescent material emits light.

(65) Because the T.sub.1 level of the host material is lower than the T.sub.1 level of the guest material, energy is transferred from T.sub.FG to T.sub.FH without deactivation of T.sub.FG (see Route D in FIG. 2B) and is utilized for TTA.

Light Emission Mechanism of First and Second Light-Emitting Layers

(66) Each light emission mechanism of the first light-emitting layer **113** and the second light-emitting layer **114** is described above. In the light-emitting element **100** of one embodiment of the present invention, even when energy is transferred from the exciplex to the host material of the first light-emitting layer **113** (in particular, when energy of the triplet excited level is transferred) at an interface between the first light-emitting layer **113** and the second light-emitting layer **114**, triplet excitation energy can be converted into light emission in the first light-emitting layer **113**.

(67) FIG. 3 shows a correlation of energy levels in the case where TTA is utilized in the first light-emitting layer **113** and ExTET is utilized in the second light-emitting layer **114**. The following

explains what terms and signs in FIG. 3 represent: Fluorescence EML: the fluorescent light-emitting layer (the first light-emitting layer **113**); Phosphorescence EML: the phosphorescent light-emitting layer (the second light-emitting layer **114**); T.sub.FH: the level of the lowest triplet excited state of the host material; S.sub.FG: the level of the lowest singlet excited state of the guest material (the fluorescent material); T.sub.FG: the level of the lowest triplet excited state of the guest material (the fluorescent material); S.sub.PH: the level of the lowest singlet excited state of the host material (the first organic compound); T.sub.PH: the level of the lowest triplet excited state of the host material (the first organic compound); T.sub.PG: the level of the lowest triplet excited state of the guest material (the phosphorescent material); S.sub.E: the level of the lowest singlet excited state of the exciplex; and T.sub.E: the level of the lowest triplet excited state of the exciplex.

(68) As shown in FIG. 3, the exciplex exists only in an excited state; thus, exciton diffusion between the exciplexes is not likely to occur. In addition, because the excited levels of the exciplex (S.sub.E and T.sub.E) are lower than the excited levels of the first organic compound (the host material of the phosphorescent material) of the second light-emitting layer **114** (S.sub.PH and T.sub.PH), energy diffusion from the exciplex to the first organic compound does not occur. That is, emission efficiency of the phosphorescent light-emitting layer (the second light-emitting layer **114**) can be maintained because an exciton diffusion distance of the exciplex is short in the phosphorescent light-emitting layer (the second light-emitting layer **114**). In addition, even when part of the triplet excitation energy of the exciplex of the phosphorescent light-emitting layer (the second light-emitting layer **114**) diffuses into the fluorescent light-emitting layer (the first light-emitting layer **113**) through the interface between the fluorescent light-emitting layer (the first light-emitting layer **113**) and the phosphorescent light-emitting layer (the second light-emitting layer **114**), energy loss can be reduced because the triplet excitation energy in the fluorescent light-emitting layer (the first light-emitting layer **113**) caused by the diffusion is used for light emission through TTA.

(69) The light-emitting element of one embodiment of the present invention can have emission efficiency exceeding the exciton generation probability when ExTET is utilized in the second light-emitting layer **114** and TTA is utilized in the first light-emitting layer **113** as described above. Thus, a light-emitting element with high efficiency can be provided.

(70) Note that in FIG. 1A, the first light-emitting layer **113** is on the side of the first electrode **101** functioning as the anode and the second light-emitting layer **114** is on the side of the second electrode **102** functioning as the cathode. However, the stacking order may be reversed. Specifically, as shown by the light-emitting element **140** in FIG. 1B, the first light-emitting layer **113** and the second light-emitting layer **114** may be on the side of the electrode functioning as the cathode and on the side of the electrode functioning as the anode, respectively. In other words, the first light-emitting layer **113** is over the second light-emitting layer **114** in the light-emitting element **140**.

(71) Such a structure used for the light-emitting element **140** is preferable when a microcavity structure (described later) is employed because the optical path length of the second light-emitting layer **114** and/or the first light-emitting layer **113** is easily adjusted.

(72) The details of the components of the light-emitting element **100** of one embodiment of the present invention are described below.

Electrode

(73) The first electrode **101** and the second electrode **102** have functions of injecting holes and electrons, respectively, into the first light-emitting layer **113** and the second light-emitting layer **114**. These electrodes can be formed of a metal, an alloy, or a conductive compound, or a mixture or a stack thereof, for example A typical example of the metal is aluminum, besides, a transition metal such as silver, tungsten, chromium, molybdenum, copper, or titanium, an alkali metal such as lithium or cesium, or a Group 2 metal such as calcium or magnesium can be used. As the transition

metal, a rare earth metal may be used. An alloy containing any of the above metals can be used as the alloy, and MgAg and AlLi can be given as examples. As the conductive compound, a metal oxide such as indium oxide-tin oxide (indium tin oxide) can be given. It is also possible to use an inorganic carbon-based material such as graphene as the conductive compound. As described above, the first electrode **101** and/or the second electrode **102** may be formed by stacking two or more of these materials.

(74) Light emitted from the first light-emitting layer **113** and the second light-emitting layer **114** is extracted through the first electrode **101** and/or the second electrode **102**. Therefore, at least one of the electrodes transmits visible light. In the case where the electrode through which light is extracted is formed using a material with low light permeability, such as metal or alloy, the first electrode **101**, the second electrode **102**, or part thereof is formed to a thickness that is thin enough to transmit visible light. In this case, the specific thickness is in a range from 1 nm to 10 nm.

First Light-Emitting Layer

(75) The first light-emitting layer **113** includes the host material and the fluorescent material. In the first light-emitting layer **113**, the host material is present in the highest proportion by weight, and the fluorescent material is dispersed in the host material. The S.sub.1 level of the host material is higher than the S.sub.1 level of the fluorescent material, and the T.sub.1 level of the host material is lower than the T.sub.1 level of the fluorescent material.

(76) An anthracene derivative or a tetracene derivative is preferably used as the host material. This is because these derivatives each have a high S.sub.1 level and a low T.sub.1 level. Specific examples include 9-phenyl-3-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazole (PCzPA), 3-[4-(1-naphthyl)-phenyl]-9-phenyl-9H-carbazole (PCPN), 9-[4-(10-phenyl-9-anthracenyl)phenyl]-9H-carbazole (CzPA), 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (cgDBCzPA), 6-[3-(9,10-diphenyl-2-anthryl)phenyl]-benzo[b]naphtho[1,2-d]furan (2mBnfPPA), and 9-phenyl-10-{4-(9-phenyl-9H-fluoren-9-yl)biphenyl-4'-yl}anthracene (FLPPA). Besides, 5,12-diphenyltetracene, 5,12-bis(biphenyl-2-yl)tetracene, and the like can be given.

(77) Examples of the fluorescent material include a pyrene derivative, an anthracene derivative, a triphenylene derivative, a fluorene derivative, a carbazole derivative, a dibenzothiophene derivative, a dibenzofuran derivative, a dibenzoquinoxaline derivative, a quinoxaline derivative, a pyridine derivative, a pyrimidine derivative, a phenanthrene derivative, and a naphthalene derivative. A pyrene derivative is particularly preferable because it has a high emission quantum yield. Specific examples of the pyrene derivative include N,N'-bis(3-methylphenyl)-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (1,6mMemFLPAPrn), N,N'-bis[4-(9-phenyl-9H-fluoren-9-yl)phenyl]-N,N'-diphenylpyrene-1,6-diamine (1,6FLPAPrn), N,N'-bis(dibenzofuran-2-yl)-N,N'-diphenylpyrene-1,6-diamine (1,6FrAPrn), and N,N'-bis(dibenzothiophen-2-yl)-N,N'-diphenylpyrene-1,6-diamine (1,6ThAPrn).

Second Light-Emitting Layer

(78) The second light-emitting layer **114** includes the first organic compound, the second organic compound, and the phosphorescent material. Note that the first organic compound serves as the host material and the second organic compound serves as the assist material in the following description.

(79) In the second light-emitting layer **114**, the host material (the first organic compound) is present in the highest proportion by weight, and the phosphorescent material is dispersed in the host material. The T.sub.1 level of the host material (the first organic compound) of the second light-emitting layer **114** is preferably higher than the T.sub.1 level of the fluorescent material of the first light-emitting layer **113**.

(80) As the phosphorescent material, an iridium-, rhodium-, or platinum-based organometallic complex or metal complex can be used; in particular, an organoiridium complex such as an iridium-based ortho-metalated complex is preferable. As an ortho-metalated ligand, a 4H-triazole ligand, a 1H-triazole ligand, an imidazole ligand, a pyridine ligand, a pyrimidine ligand, a pyrazine ligand,

an isoquinoline ligand, or the like can be given. As the metal complex, a platinum complex having a porphyrin ligand or the like can be given.

(81) As the phosphorescent material, a material with a spectrum peak in the yellow wavelength region is preferred. In addition, it is preferred that an emission spectrum of the material with the peak in the yellow wavelength region include spectral components in the green and red wavelength regions.

(82) Examples of the host material (the first organic compound) include a zinc- or aluminum-based metal complex, an oxadiazole derivative, a triazole derivative, a benzimidazole derivative, a quinoxaline derivative, a dibenzoquinoxaline derivative, a dibenzothiophene derivative, a dibenzofuran derivative, a pyrimidine derivative, a triazine derivative, a pyridine derivative, a bipyridine derivative, and a phenanthroline derivative. Other examples are an aromatic amine and a carbazole derivative.

(83) As the second organic compound (the assist material), a substance which can form an exciplex together with the first organic compound is used. In this case, it is preferable that the first organic compound, the second organic compound, and the phosphorescent material be selected such that the emission peak of the exciplex overlaps with an adsorption band, specifically an adsorption band on the longest wavelength side, of a triplet metal to ligand charge transfer (MLCT) transition of the phosphorescent material. This makes it possible to provide a light-emitting element with drastically improved emission efficiency. However, if a material exhibiting thermally activated delayed fluorescence (TADF) is used instead of the phosphorescent material, it is preferred that an adsorption band on the longest wavelength side be an absorption band of a singlet. The TADF material is explained later.

(84) There is no limitation on the emission colors of the first light-emitting material and the second light-emitting material, and they may be the same or different. Light emitted from the light-emitting materials is mixed and extracted out of the element; therefore, for example, in the case where their emission colors are complementary colors, the light-emitting element can emit white light. In consideration of the reliability of the light-emitting element, the emission peak wavelength of the first light-emitting material is preferably shorter than that of the second light-emitting material. For example, it is preferable that the first light-emitting material emit blue light and the second light-emitting material emit yellow light.

Other Layers

(85) As illustrated in FIG. 1A, the light-emitting element **100** of one embodiment of the present invention may include another layer besides the first light-emitting layer **113** and the second light-emitting layer **114**. For example, the light-emitting element may include a hole-injection layer, a hole-transport layer, an electron-blocking layer, a hole-blocking layer, an electron-transport layer, an electron-injection layer, and the like. Furthermore, each of these layers may be formed of a plurality of layers. These layers can reduce a carrier injection barrier, improve a carrier transport property, or suppress a quenching phenomenon by an electrode, thereby contributing to an improvement in emission efficiency or a reduction in a driving voltage. Each of these layers can be formed by any one or any combination of the following methods: an evaporation method (including a vacuum evaporation method), a printing method (such as relief printing, intaglio printing, gravure printing, planography printing, and stencil printing), an ink-jet method, a coating method, and the like.

Hole-Injection Layer

(86) The hole-injection layer **111** has a function of reducing a barrier for hole injection from the first electrode **101** to promote hole injection and is formed using a transition metal oxide, a phthalocyanine derivative, or an aromatic amine, for example. As the transition metal oxide, molybdenum oxide, vanadium oxide, ruthenium oxide, tungsten oxide, manganese oxide, or the like can be given. As the phthalocyanine derivative, phthalocyanine, metal phthalocyanine, or the like can be given. As the aromatic amine, a benzidine derivative, a phenylenediamine derivative, or

the like can be given. It is also possible to use a high molecular compound such as polythiophene or poly aniline; a typical example thereof is poly(ethylenedioxythiophene)/poly(styrenesulfonic acid), which is self-doped polythiophene.

(87) As the hole-injection layer **111**, a composite material of a hole-transport material and a material having a property of accepting electrons from the hole-transport material can also be used. Alternatively, a stack of a layer containing a material having an electron accepting property and a layer containing a hole-transport material may also be used. In a steady state or in the presence of an electric field, electric charge can be transferred between these materials. As examples of the material having an electron accepting property, organic acceptors such as a quinodimethane derivative, a chloranil derivative, and a hexaazatriphenylene derivative can be given. Alternatively, a transition metal oxide such as an oxide of a metal from Group 4 to Group 8 can also be used. Specifically, vanadium oxide, niobium oxide, tantalum oxide, chromium oxide, molybdenum oxide, tungsten oxide, manganese oxide, rhenium oxide, or the like can be used. In particular, molybdenum oxide is preferable because it is stable in the air, has a low hygroscopic property, and is easily handled.

(88) A material having a property of transporting more holes than electrons can be used as the hole-transport material, and a material having a hole mobility of $1 \times 10^{-6} \text{ cm}^2/\text{Vs}$ or higher is preferable. Specifically, an aromatic amine, a carbazole derivative, an aromatic hydrocarbon, a stilbene derivative, or the like can be used. Furthermore, the hole-transport material may be a high molecular compound.

Hole-Transport Layer

(89) The hole-transport layer **112** is a layer containing a hole-transport material and can be formed using any of the materials given as examples of the material of the hole-injection layer **111**. In order that the hole-transport layer **112** has a function of transporting holes injected into the hole-injection layer **111** to the first light-emitting layer **113**, the highest occupied molecular orbital (HOMO) level of the hole-transport layer **112** is preferably equal or close to the HOMO level of the hole-injection layer **111**.

Electron-Transport Layer

(90) The electron-transport layer **115** has a function of transporting, to the second light-emitting layer **114**, electrons injected from the second electrode **102** through the electron-injection layer **116**. A material having a property of transporting more electrons than holes can be used as an electron-transport material, and a material having an electron mobility of $1 \times 10^{-6} \text{ cm}^2/\text{Vs}$ or higher is preferable. Specific examples include a metal complex having a quinoline ligand, a benzoquinoline ligand, an oxazole ligand, or a thiazole ligand; an oxadiazole derivative; a triazole derivative; a phenanthroline derivative; a pyridine derivative; and a bipyridine derivative.

Electron-Injection Layer

(91) The electron-injection layer **116** has a function of reducing a barrier for electron injection from the second electrode **102** to promote electron injection and can be formed using a Group 1 metal or a Group 2 metal, or an oxide, a halide, or a carbonate of any of the metals, for example. Alternatively, a composite material containing an electron-transport material (described above) and a material having a property of donating electrons to the electron-transport material can also be used. As the material having an electron donating property, a Group 1 metal, a Group 2 metal, an oxide of any of the metals, or the like can be given.

Substrate, FET, and the Like

(92) The light-emitting element **100** is fabricated over a substrate of glass, plastic, or the like. As the way of stacking layers over the substrate, layers may be sequentially stacked from the first electrode **101** side or sequentially stacked from the second electrode **102** side. The light-emitting element may be formed over an electrode electrically connected to a field-effect transistor (FET), for example, that is formed over a substrate of glass, plastic, or the like. Accordingly, an active matrix light-emitting device in which the FET controls the drive of the light-emitting element can

be fabricated.

(93) Although the light-emitting material included in the second light-emitting layer **114** is the phosphorescent material in the above description, the light-emitting material is not limited thereto. As the light-emitting material included in the second light-emitting layer **114**, any material can be used as long as the material can convert the triplet excitation energy into light emission. As an example of the material that can convert the triplet excitation energy into light emission, a TADF material is given in addition to a phosphorescent material. Therefore, it is acceptable that the “phosphorescent material” in the description is replaced with the “TADF material. Note that the TADF material is a substance that can up-convert a triplet excited state into a singlet excited state (i.e., reverse intersystem crossing is possible) using a little thermal energy and efficiently exhibits light emission (fluorescence) from the singlet excited state. The TADF is efficiently obtained under the condition where the difference in energy between the triplet excited level and the singlet excited level is greater than or equal to 0 eV and less than or equal to 0.2 eV, preferably greater than or equal to 0 eV and less than or equal to 0.1 eV.

(94) It is to be noted that this embodiment can be combined appropriately with other embodiments. Embodiment 2

(95) Light-emitting elements with different structures from those of the light-emitting elements **100** and **140** shown in Embodiment 1 are described in this embodiment with reference to FIGS. **4A** and **4B**. FIG. **4A** is a schematic cross-sectional view of a light-emitting element **150** of one embodiment of the present invention, and FIG. **4B** is a schematic cross-sectional view of a light-emitting element **160** of one embodiment of the present invention.

(96) The light-emitting element **150** is different from the light-emitting element **100** in that a separation layer **120** is provided between the first light-emitting layer **113** and the second light-emitting layer **114**. The separation layer **120** is in contact with the first light-emitting layer **113** and the second light-emitting layer **114**. The structures of the other layers are similar to those in Embodiment 1; therefore, description thereof is omitted.

(97) The separation layer **120** is provided to prevent energy transfer by the Dexter mechanism (particularly triplet energy transfer) from the first organic compound in an excited state or the phosphorescent material in an excited state which is generated in the second light-emitting layer **114** to the host material or the fluorescent material in the first light-emitting layer **113**. Therefore, the thickness of the separation layer may be approximately several nanometers, specifically 0.1 nm or more and 20 nm or less, 1 nm or more and 10 nm or less, or 1 nm or more and 5 nm or less.

(98) The separation layer **120** may contain a single material or both a hole-transport material and an electron-transport material. In the case of a single material, a bipolar material may be used. The bipolar material here refers to a material in which the ratio between the electron mobility and the hole mobility is 100 or less. As a material contained in the separation layer **120**, the hole-transport material, the electron-transport material, or the like given as an example in Embodiment 1 can be used. Furthermore, at least one of materials contained in the separation layer **120** may be the same as the host material (the first organic compound) of the second light-emitting layer **114**. This facilitates the manufacture of the light-emitting element and reduces the driving voltage.

(99) For example, when the separation layer **120** is formed of the same materials as the host material (the first organic compound) and the assist material (the second organic compound) of the second light-emitting layer **114**, the first light-emitting layer **113** and the second light-emitting layer **114** are stacked with each other while the layer (the separation layer **120**) not including the phosphorescent material of the second light-emitting layer **114** is provided therebetween. In the case of such a structure, depending on using or not using the phosphorescent material, the second light-emitting layer **114** or the separation layer **120** can be deposited. In other words, the separation layer **120** includes a region not including the phosphorescent material while the second light-emitting layer **114** includes a region including the phosphorescent material. In the case of such a structure, the separation layer **120** and the second light-emitting layer **114** can be formed in the

same chamber. Thus, the manufacturing cost can be reduced.

(100) Alternatively, at least one of materials contained in the separation layer **120** may have a higher T.sub.1 level than the host material (the first organic compound) of the second light-emitting layer **114**.

(101) The recombination region can be adjusted by adjusting the mixture ratio of the hole-transport material and the electron-transport material, whereby the emission color can be controlled. For example, in the case where the first electrode **101** and the second electrode **102** serve as an anode and a cathode, respectively, the recombination region can be shifted from the first electrode **101** side to the second electrode **102** side by increasing the proportion of the hole-transport material in the separation layer **120**. As a result, the contribution of the second light-emitting layer **114** to light emission can be increased. In contrast, by increasing the proportion of the electron-transport material, the recombination region can be shifted from the second electrode **102** side to the first electrode **101** side, so that the contribution of the first light-emitting layer **113** to light emission can be increased. In the case where the first light-emitting layer **113** and the second light-emitting layer **114** have different emission colors, the emission color of the light-emitting element can be changed as a whole.

(102) The hole-transport material and the electron-transport material may form an exciplex in the separation layer **120**, which effectively prevents exciton diffusion. Specifically, energy transfer from the host material (the first organic compound) of the second light-emitting layer **114** in an excited state or the phosphorescent material in an excited state to the host material of the first light-emitting layer **113** or the fluorescent material can be prevented.

(103) As in the light-emitting element **140** described in Embodiment 1, the first light-emitting layer **113** may be positioned over the second light-emitting layer **114**. Specifically, as shown by the light-emitting element **160** in FIG. 4B, it is acceptable that the second light-emitting layer **114** is provided over the hole-transport layer **112**, and the first light-emitting layer **113** is provided over the second light-emitting layer **114** with the separation layer **120** provided therebetween.

(104) The structure described in this embodiment can be used in appropriate combination with any of the structures described in the other embodiments.

Embodiment 3

(105) In this embodiment, a light-emitting element of one embodiment of the present invention is described with reference to FIG. 5. FIG. 5 is a schematic cross-sectional view of a light-emitting element **170** of one embodiment of the present invention.

(106) The light-emitting element **170** includes a plurality of light-emitting units (a first light-emitting unit **131** and a second light-emitting unit **132** in FIG. 5) between the first electrode **101** and the second electrode **102**. One light-emitting unit has the same structure as the EL layer **130** illustrated in FIG. 1A or 1B. That is, the light-emitting element **100** in FIG. 1A includes one light-emitting unit while the light-emitting element **170** includes the plurality of light-emitting units.

(107) In the light-emitting element **170** shown in FIG. 5, the first light-emitting unit **131** and the second light-emitting unit **132** are stacked, and a charge generation layer **133** is provided between the first light-emitting unit **131** and the second light-emitting unit **132**. Note that the first light-emitting unit **131** and the second light-emitting unit **132** may have the same structure or different structures.

(108) The charge generation layer **133** may include a composite material of an organic compound and metal oxide. For the composite material, the composite material that can be used for the hole-injection layer **111** described above may be used. As the organic compound, a variety of compounds such as an aromatic amine compound, a carbazole compound, an aromatic hydrocarbon, and a high molecular compound (such as an oligomer, a dendrimer, or a polymer) can be used. An organic compound having a hole mobility of 1×10^{-6} cm²/Vs or higher is preferably used. Note that any other substance may be used as long as the substance has a hole-transport property higher than an electron-transport property. Since the composite material of an

organic compound and a metal oxide is superior in carrier-injecting property and carrier-transporting property, low-voltage driving or low-current driving can be realized. Note that when a surface of a light-emitting unit on the anode side is in contact with the charge generation layer **133**, the charge generation layer **133** can also serve as a hole-transport layer of the light-emitting unit; thus, a hole-transport layer does not need to be formed in the light-emitting unit.

(109) The charge generation layer **133** may have a stacked-layer structure of a layer containing the composite material of an organic compound and a metal oxide and a layer containing another material. For example, the charge generation layer **133** may be formed using a combination of a layer containing the composite material of an organic compound and a metal oxide with a layer containing one compound selected from among electron-donating substances and a compound having a high electron-transporting property. Further, the charge generation layer **133** may be formed using a combination of a layer containing the composite material of an organic compound and a metal oxide with a transparent conductive film.

(110) In any case, as the charge-generation layer **133**, which is provided between the first light-emitting unit **131** and the second light-emitting unit **132**, acceptable is a layer which injects electrons into the light-emitting unit on one side and injects holes into the light-emitting unit on the other side when voltage is applied to the first electrode **101** and the second electrode **102**. For example, in FIG. 5, when a voltage is applied such that a potential of the first electrode **101** is higher than a potential of the second electrode **102**, any structure may be used for the charge generation layer **133**, as long as the charge generation layer **133** injects electrons and holes into the first light-emitting unit **131** and the second light-emitting unit **132**, respectively.

(111) In FIG. 5, the light-emitting element having two light-emitting units is described; however, one embodiment of the present invention can be similarly applied to a light-emitting element in which three or more light-emitting units are stacked. With a plurality of light-emitting units partitioned by the charge generation layer between a pair of electrodes as in the light-emitting element **170**, it is possible to provide a light-emitting element which can emit light with high luminance with the current density kept low and has a long lifetime. A light-emitting device that can be driven at a low voltage and has low power consumption can be realized.

(112) When the above-described structure of the EL layer **130** is used for at least one of the plurality of units, the number of manufacturing steps of the unit can be reduced; thus, a multicolor light-emitting element which is advantageous for practical application can be provided.

(113) The above-described structure can be combined with any of the structures in this embodiment and the other embodiments.

Embodiment 4

(114) In this embodiment, a light-emitting device manufactured using the light-emitting element described in any of Embodiments 1 to 3 will be described with reference to FIGS. **6A** and **6B**.

(115) FIG. **6A** is a top view illustrating a light-emitting device **600** and FIG. **6B** is a cross-sectional view taken along the dashed-dotted line A-B and the dashed-dotted line C-D in FIG. **6A**. The light-emitting device **600** includes driver circuit portions (a source line driver circuit portion **601** and a gate line driver circuit portion **603**) and a pixel portion **602**. Note that the source line driver circuit portion **601**, the gate line driver circuit portion **603**, and the pixel portion **602** have a function of controlling light emission of a light-emitting element.

(116) The light-emitting device **600** also includes an element substrate **610**, a sealing substrate **604**, a sealing member **605**, a region **607** surrounded by the sealing member **605**, a lead wiring **608**, and an FPC **609**.

(117) Note that the lead wiring **608** is a wiring for transmitting signals to be input to the source line driver circuit portion **601** and the gate line driver circuit portion **603** and for receiving a video signal, a clock signal, a start signal, a reset signal, and the like from the FPC **609** serving as an external input terminal. Although only the FPC **609** is shown here, the FPC **609** may be provided with a printed wiring board (PWB).

(118) In the source line driver circuit portion **601**, a CMOS circuit is formed in which an re-channel FET **623** and a p-channel FET **624** are combined. Note that the source line driver circuit portion **601** or the gate line driver circuit portion **603** may be formed with various kinds of CMOS circuits, NMOS circuits, and PMOS circuits. In this embodiment, although a driver-integrated type structure in which a driver circuit portion is formed over a substrate is described, a driver circuit portion is not necessarily formed over a substrate but can be formed outside a substrate.

(119) The pixel portion **602** includes a switching FET **611**, a current control FET **612**, and a first electrode **613** electrically connected to a drain of the current control FET **612**. It is to be noted that an insulator **614** is formed to cover an edge of the first electrode **613**. As the insulator **614**, for example, a positive type photosensitive acrylic resin film can be used.

(120) The insulator **614** is formed to have a curved surface with curvature at an upper edge or a lower edge thereof in order to obtain favorable coverage. For example, in the case where positive photosensitive acrylic is used for a material of the insulator **614**, it is preferred that only the upper end portion of the insulator **614** has a curved surface with a curvature radius (0.2 μm to 3 μm). As the insulator **614**, either a negative photosensitive resin or a positive photosensitive resin can be used.

(121) Note that there is no particular limitation on a structure of each of the FETs (the FETs **611**, **612**, **623**, and **624**). For example, a staggered transistor can be used. In addition, there is no particular limitation on a conductivity type of each transistor. For these transistors, n-type and p-type transistors may be used, or either n-type transistors or p-type transistors may be used, for example. Furthermore, there is no particular limitation on crystallinity of a semiconductor film used for the transistor. For example, an amorphous semiconductor film or a crystalline semiconductor film may be used. Examples of a semiconductor material include Group 13 semiconductors (e.g., gallium), Group 14 semiconductors (e.g., silicon), compound semiconductors (including oxide semiconductors), organic semiconductors, and the like. For example, an oxide semiconductor that has an energy gap of 2 eV or more, preferably 2.5 eV or more, further preferably 3 eV or more is preferably used for the transistors, so that the off-state current of the transistors can be reduced. Examples of the oxide semiconductor include an In—Ga oxide, an In—M—Zn oxide (M is Al, Ga, Y, Zr, La, Ce, or Nd), and the like.

(122) An EL layer **616** and a second electrode **617** are formed over the first electrode **613**. Here, the first electrode **613** serves as an anode and the second electrode **617** serves as a cathode.

(123) The EL layer **616** can be formed by a method such as an evaporation method (including a vacuum evaporation method), a printing method (such as relief printing, intaglio printing, gravure printing, planography printing, and stencil printing), an ink jet method, or a coating method. The EL layer **616** has the structure described in any of Embodiments 1 to 3. As another material included in the EL layer **616**, a low molecular compound or a high molecular compound (including an oligomer or a dendrimer) may be used.

(124) Note that the light-emitting element **618** is formed with the first electrode **613**, the EL layer **616**, and the second electrode **617**. The light-emitting element **618** has any of the structures shown in Embodiments 1 to 3. In the case where the pixel portion includes a plurality of light-emitting elements, the pixel portion may include both the light-emitting element described in any of Embodiments 1 to 3 and a light-emitting element having a different structure.

(125) When the sealing substrate **604** and the element substrate **610** are attached to each other with the sealing member **605**, the light-emitting element **618** is provided in the region **607** surrounded by the element substrate **610**, the sealing substrate **604**, and the sealing member **605**. Note that the region **607** is filled with filler, specifically filled with an inert gas (such as nitrogen or argon) in some cases, or filled with the sealing member **605** in other cases. It is preferable that the sealing substrate be provided with a recessed portion and the drying agent (not illustrated in the drawing) be provided in the recessed portion, in which case deterioration due to influence of moisture can be suppressed.

(126) An epoxy-based resin or glass frit is preferably used for the sealing member **605**. The material preferably allows as little moisture and oxygen as possible to penetrate. As the sealing substrate **604**, a glass substrate, a quartz substrate, or a plastic substrate formed of fiber reinforced plastic (FRP), poly(vinyl fluoride) (PVF), polyester, acrylic, or the like can be used.

(127) As described above, the light-emitting device which uses the light-emitting element described in any of Embodiments 1 to 3 can be obtained.

(128) The light-emitting device **600** in this embodiment is fabricated using the light-emitting element described in any of Embodiments 1 to 3 and thus can have favorable characteristics. Specifically, since the light-emitting element described in any of Embodiments 1 to 3 has high emission efficiency, the light-emitting device can have reduced power consumption. In addition, since the light-emitting element is easy to mass-produce, the light-emitting device can be provided at low cost.

(129) FIGS. 7A and 7B each illustrate an example of a cross-sectional view of a light-emitting device in which full color display is achieved by formation of a light-emitting element exhibiting white light emission and use of coloring layers (color filters) and the like.

(130) In FIG. 7A, a substrate **1001**, a base insulating film **1002**, a gate insulating film **1003**, gate electrodes **1006**, **1007**, and **1008**, a first interlayer insulating film **1020**, a second interlayer insulating film **1021**, a peripheral portion **1042**, a pixel portion **1040**, a driver circuit portion **1041**, first electrodes **1024Y**, **1024R**, **1024G** and **1024B** of light-emitting elements, a partition **1025**, an EL layer **1028**, a second electrode **1026** of the light-emitting elements, a sealing substrate **1031**, a sealing member **1032**, and the like are illustrated.

(131) In FIG. 7A, coloring layers (a red coloring layer **1034R**, a green coloring layer **1034G**, a blue coloring layer **1034B**, and a yellow coloring layer **1034Y**) are provided on a transparent base material **1033**. Further, a black layer (black matrix) **1035** may be provided. The transparent base material **1033** provided with the coloring layers and the black layer is positioned and fixed to the substrate **1001**. Note that the coloring layers and the black layer are covered with an overcoat layer **1036**. In the structure in FIG. 7A, red light, blue light, green light, and yellow light transmit the coloring layers, and thus an image can be displayed with the use of pixels of four colors.

(132) FIG. 7B illustrates an example in which coloring layers (the red coloring layer **1034R**, the green coloring layer **1034G**, and the blue coloring layer **1034B**) are formed between the gate insulating film **1003** and the first interlayer insulating film **1020**. As in the structure, the coloring layers may be provided between the substrate **1001** and the sealing substrate **1031**. Note that the yellow coloring layer is not necessarily provided as shown in FIG. 7B because the light-emitting element of one embodiment of the present invention can emit light with a yellow wavelength.

(133) The above-described light-emitting device is a light-emitting device having a structure in which light is extracted from the substrate **1001** side where the FETs are formed (a bottom emission structure), but a light-emitting device having a structure in which light is extracted from the sealing substrate **1031** side (a top emission structure) is also acceptable.

(134) FIG. 8 is a cross-sectional view of a light-emitting device having a top emission structure. In this case, as the substrate **1001**, a substrate that does not transmit light can be used. The process up to the step of forming a connection electrode which connects the FET and the anode of the light-emitting element is performed in a manner similar to that of the light-emitting device having a bottom emission structure. Then, a third interlayer insulating film **1037** is formed to cover an electrode **1022**. This insulating film may function for planarization. The third interlayer insulating film **1037** can be formed by using a material similar to that of the second interlayer insulating film, or can be formed by using any other materials.

(135) The first electrodes **1024Y**, **1024R**, **1024G**, and **1024B** of the light-emitting elements each serve as an anode here, but may serve as a cathode. Further, in the case of a light-emitting device having a top emission structure as illustrated in FIG. 8, the first electrodes **1024Y**, **1024R**, **1024G**; and **1024B** are preferably reflective electrodes. The EL layer **1028** is formed to have a structure

similar to the structure of the EL layer **130**, which is described in any of Embodiments 1 to 3, with which white light emission can be obtained.

(136) In FIG. **8**, a second electrode **1026** is formed over the EL layer **1028**. The second electrode **1026** may be a semi-transmissive and semi-reflective electrode, and a micro optical resonator (microcavity) structure utilizing a resonant effect of light between the second electrode **1026** and the first electrodes **1024Y**, **1024R**, **1024G**, and **1024B** may be used so as to increase the intensity of light having a specific wavelength.

(137) In the case of a top emission structure as illustrated in FIG. **8**, sealing can be performed with the sealing substrate **1031** on which the coloring layers (the red coloring layer **1034R**, the green coloring layer **1034G**, the blue coloring layer **1034B**, and the yellow coloring layer **1034Y**) are provided. The sealing substrate **1031** may be provided with a black layer **1035** which is positioned between pixels. The color layers (the red coloring layer **1034R**, the green coloring layer **1034G** and the blue coloring layer **1034B**, and the yellow coloring layer **1034Y**) and the black layer **1035** may be covered with the overcoat layer (not illustrated in the drawing). Note that a light-transmitting substrate is used as the sealing substrate **1031**.

(138) FIG. **8** shows the structure provided with the light-emitting elements from which white light emission can be obtained and the coloring layers for the light-emitting elements as an example; however, the structure is not limited thereto. For example, as shown in FIG. **9A**, a structure including the light-emitting elements from which white light emission can be obtained, the red coloring layer **1034R**, the green coloring layer **1034G** and the yellow coloring layer **1034Y** while not including the blue coloring layer may be employed in order to achieve full color display with the four colors of red, green, blue, and yellow. Alternatively, as shown in FIG. **9B**, a structure including the light-emitting elements from which white light emission can be obtained, the red coloring layer **1034R**, and the green coloring layer **1034G** while not including the blue coloring layer and the yellow coloring layer may be employed in order to achieve full color display with the four colors of red, green, blue, and yellow. The structure as shown in FIG. **8** where the coloring layer is provided to each of the light-emitting element from which white light emission can be obtained is effective to suppress reflection of outside light. In contrast, the structure as shown in FIG. **9B** where the light-emitting elements from which white light emission can be obtained are provided with the red coloring layer and the green coloring layer and without the blue and yellow coloring layers is effective to reduce power consumption because of small energy loss of light emitted from the light-emitting elements.

(139) Alternatively, a structure as shown in FIG. **10A**, including the light-emitting elements from which white light emission can be obtained and the coloring layers for the light-emitting layers to achieve full color display with three colors of red, green, and blue, may be employed. Alternatively, a structure as shown in FIG. **10B**, including the light-emitting elements from which white light emission can be obtained, the red coloring layer **1034R**, and the green coloring layer **1034G** while not including the blue coloring layer to achieve full color display with three colors of red, green, and blue, may be employed.

(140) Note that FIGS. **9A** and **9B** and FIGS. **10A** and **10B** are each a schematic cross-sectional view for illustrating the light-emitting device of one embodiment of the present invention, and the driver circuit portion **1041**, the peripheral portion **1042**, and the like, which are shown in FIG. **8**, are not illustrated therein. In the structures of FIGS. **9A** and **9B** and FIGS. **10A** and **10B**, the microcavity structure may be employed as in the structure of FIG. **8**.

(141) The light-emitting device in this embodiment is fabricated using the light-emitting element described in any of Embodiments 1 to 3 and thus can have favorable characteristics. Specifically, since the light-emitting element described in any of Embodiments 1 to 3 has high emission efficiency, the light-emitting device can have reduced power consumption. When the light-emitting element described in any of Embodiments 1 to 3 is combined with the coloring layer such as the color filter, an optimum element structure can be formed from which white light emission can be

obtained. In addition, since the structure of the light-emitting element described in any of Embodiments 1 to 3 is easy to mass-produce, the light-emitting device can be provided at low cost. (142) The above-described structure can be combined with any of the structures in this embodiment and the other embodiments.

Embodiment 5

(143) In this embodiment, a display device that includes a light-emitting device of one embodiment of the present invention is described with reference to FIGS. **11A** and **11B**.

(144) FIG. **11A** is a block diagram illustrating the display device of one embodiment of the present invention, and FIG. **11B** is a circuit diagram illustrating a pixel circuit of the display device of one embodiment of the present invention.

(145) The display device illustrated in FIG. **11A** includes a region including pixels of display elements (hereinafter the region is referred to as a pixel portion **802**), a circuit portion provided outside the pixel portion **802** and including a circuit for driving the pixels (hereinafter the portion is referred to as a driver circuit portion **804**), circuits having a function of protecting elements (hereinafter the circuits are referred to as protection circuits **806**), and a terminal portion **807**. Note that the protection circuits **806** are not necessarily provided.

(146) A part or the whole of the driver circuit portion **804** is preferably formed over a substrate over which the pixel portion **802** is formed. Thus, the number of components and the number of terminals can be reduced. When a part or the whole of the driver circuit portion **804** is not formed over the substrate over which the pixel portion **802** is formed, the part or the whole of the driver circuit portion **804** can be mounted by COG or tape automated bonding (TAB).

(147) The pixel portion **802** includes circuits for driving a plurality of display elements arranged in X rows (X is a natural number of 2 or more) and Y columns (Y is a natural number of 2 or more) (hereinafter, such circuits are referred to as pixel circuits **801**). The driver circuit portion **804** includes driver circuits such as a circuit for supplying a signal (scan signal) to select a pixel (hereinafter the circuit is referred to as a gate driver **804a**) and a circuit for supplying a signal (data signal) to drive a display element in a pixel (hereinafter, the circuit is referred to as a source driver **804b**).

(148) The gate driver **804a** includes a shift register or the like. The gate driver **804a** receives a signal for driving the shift register through the terminal portion **807** and outputs a signal. For example, the gate driver **804a** receives a start pulse signal, a clock signal, or the like and outputs a pulse signal. The gate driver **804a** has a function of controlling the potentials of wirings supplied with scan signals (hereinafter, such wirings are referred to as scan lines GL_1 to GL_X). Note that a plurality of gate drivers **804a** may be provided to control the scan lines GL_1 to GL_X separately. Alternatively, the gate driver **804a** has a function of supplying an initialization signal. Not limited thereto, the gate driver **804a** can supply another signal.

(149) The source driver **804b** includes a shift register or the like. The source driver **804b** receives a signal (video signal) from which a data signal is derived, as well as a signal for driving the shift register, through the terminal portion **807**. The source driver **804b** has a function of generating a data signal to be written in the pixel circuits **801** based on the video signal. In addition, the source driver **804b** has a function of controlling output of a data signal in response to a pulse signal produced by input of a start pulse signal, a clock signal, or the like. Further, the source driver **804b** has a function of controlling the potentials of wirings supplied with data signals (hereinafter, such wirings are referred to as data lines DL_1 to DL_Y). Alternatively, the source driver **804b** has a function of supplying an initialization signal. Not limited thereto, the source driver **804b** can supply another signal.

(150) Alternatively, the source driver **804b** is formed using a plurality of analog switches or the like, for example. The source driver **804b** can output, as the data signals, signals obtained by time-dividing the video signal by sequentially turning on the plurality of analog switches. The source driver **804b** may include a shift register or the like.

(151) A pulse signal and a data signal are input, through one of the plurality of scan lines GL supplied with scan signals and one of the plurality of data lines DL supplied with data signals, respectively, to each of the plurality of the pixel circuits **801**. Writing and holding of the data signal in each of the plurality of pixel circuits **801** are controlled by the gate driver **804a**. For example, to the pixel circuit **801** in the m-th row and the n-th column (in is a natural number of less than or equal to X, and n is a natural number of less than or equal to Y), a pulse signal is input from the gate driver **804a** through the scan line GL_m, and a data signal is input from the source driver **804b** through the data line DL_n in accordance with the potential of the scan line GL_m.

(152) The protection circuit **806** shown in FIG. **11A** is connected to, for example, the scan line GL between the gate driver **804a** and the pixel circuits **801**. Alternatively, the protection circuit **806** is connected to the data line DL between the source driver **804b** and the pixel circuit **801**.

Alternatively, the protection circuit **806** can be connected to a wiring between the gate driver **804a** and the terminal portion **807**. Alternatively, the protection circuit **806** can be electrically connected to a wiring between the source driver **804b** and the terminal portion **807**. Note that the terminal portion **807** means a portion having terminals for inputting power, control signals, and video signals to the display device from external circuits.

(153) The protection circuit **806** is a circuit which electrically conducts a wiring connected to the protection circuit to another wiring when a potential out of a certain range is supplied to the wiring connected to the protection circuit.

(154) As illustrated in FIG. **11A**, the protection circuits **806** are provided for the pixel portion **802** and the driver circuit portion **804**, so that the resistance of the display device to overcurrent generated by electrostatic discharge (ESD) or the like can be improved. Note that the configuration of the protection circuits **806** is not limited to that, and for example, the protection circuit **806** may be configured to be connected to the gate driver **804a** or the protection circuit **806** may be configured to be connected to the source driver **804b**. Alternatively, the protection circuit **806** may be configured to be connected to the terminal portion **807**.

(155) In FIG. **11A**, an example in which the driver circuit portion **804** includes the gate driver **804a** and the source driver **804b** is shown; however, the structure is not limited thereto. For example, only the gate driver **804a** may be formed and a separately prepared substrate where a source driver circuit is formed (e.g., a driver circuit substrate formed with a single crystal semiconductor film or a polycrystalline semiconductor film) may be mounted.

(156) Each of the plurality of pixel circuits **801** in FIG. **11A** can have the structure illustrated in FIG. **11B**, for example.

(157) The pixel circuit **801** shown in FIG. **11B** includes transistors **852** and **854**, a capacitor **862**, and a light-emitting element **872**.

(158) One of a source electrode and a drain electrode of the transistor **852** is electrically connected to a wiring to which a data signal is supplied (hereinafter referred to as a signal line DL_n). A gate electrode of the transistor **852** is electrically connected to a wiring to which a gate signal is supplied (hereinafter referred to as a scan line GL_m).

(159) The transistor **852** has a function of controlling whether to write a data signal by being turned on or off.

(160) One of a pair of electrodes of the capacitor **862** is electrically connected to a wiring to which a potential is supplied (hereinafter referred to as a potential supply line VL_a), and the other is electrically connected to the other of the source electrode and the drain electrode of the transistor **852**.

(161) The capacitor **862** functions as a storage capacitor for storing written data.

(162) One of a source electrode and a drain electrode of the transistor **854** is electrically connected to the potential supply line VL_a. Further, a gate electrode of the transistor **854** is electrically connected to the other of the source electrode and the drain electrode of the transistor **852**.

(163) One of an anode and a cathode of the light-emitting element **872** is electrically connected to a

potential supply line VL_b, and the other is electrically connected to the other of the source electrode and the drain electrode of the transistor **854**.

(164) As the light-emitting element **872**, the light-emitting element described in any of Embodiments 1 to 3 can be used.

(165) A high power supply potential VDD is supplied to one of the potential supply line VL_a and the potential supply line VL_b, and a low power supply potential VSS is supplied to the other.

(166) For example, in the display device including the pixel circuit **801** in FIG. **11B**, the pixel circuits **801** are sequentially selected row by row by the gate driver **804a** illustrated in FIG. **11A**, whereby the transistor **852** is turned on and a data signal is written.

(167) When the transistor **852** is turned off, the pixel circuits **801** in which the data has been written are brought into a holding state. Further, the amount of current flowing between the source electrode and the drain electrode of the transistor **854** is controlled in accordance with the potential of the written data signal. The light-emitting element **872** emits light with a luminance corresponding to the amount of flowing current. This operation is sequentially performed row by row; thus, an image is displayed.

(168) The structure described in this embodiment can be used in appropriate combination with the structure described in any of the other embodiments.

Embodiment 6

(169) In this embodiment, a display module and electronic devices that include a light-emitting device of one embodiment of the present invention are described with reference to FIG. **12** and FIGS. **13A** to **13G**.

(170) In a display module **8000** illustrated in FIG. **12**, a touch panel **8004** connected to an FPC **8003**, a display panel **8006** connected to an FPC **8005**, a frame **8009**, a printed board **8010**, and a battery **8011** are provided between an upper cover **8001** and a lower cover **8002**.

(171) The light-emitting device of one embodiment of the present invention can be used for, for example, the display panel **8006**.

(172) The shapes and sizes of the upper cover **8001** and the lower cover **8002** can be changed as appropriate in accordance with the sizes of the touch panel **8004** and the display panel **8006**.

(173) The touch panel **8004** can be a resistive touch panel or a capacitive touch panel and may overlap with the display panel **8006**. Alternatively, a counter substrate (sealing substrate) of the display panel **8006** can have a touch panel function. Alternatively, a photosensor may be provided in each pixel of the display panel **8006** so as to function as an optical touch panel.

(174) The frame **8009** protects the display panel **8006** and functions as an electromagnetic shield for blocking electromagnetic waves generated by the operation of the printed board **8010**. The frame **8009** can function as a radiator plate.

(175) The printed board **8010** is provided with a power supply circuit and a signal processing circuit for outputting a video signal and a clock signal. As a power source for supplying power to the power supply circuit, an external commercial power source or a power source using the battery **8011** provided separately may be used. The battery **8011** can be omitted in the case of using a commercial power source.

(176) The display module **8000** may be additionally provided with a member such as a polarizing plate, a retardation plate, or a prism sheet.

(177) FIGS. **13A** to **13G** illustrate electronic devices. These electronic devices can include a housing **9000**, a display portion **9001**, a speaker **9003**, operation keys **9005** (including a power switch or an operation switch), a connection terminal **9006**, a sensor **9007** (a sensor having a function of measuring or sensing force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, chemical substance, sound, time, hardness, electric field, current, voltage, electric power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared ray), a microphone **9008**, and the like.

(178) The electronic devices illustrated in FIGS. **13A** to **13G** can have a variety of functions. For

example, a function of displaying a variety of data (a still image, a moving image, a text image, and the like) on the display portion, a touch sensor function, a function of displaying a calendar, date, time, and the like, a function of controlling a process with a variety of software (programs), a wireless communication function, a function of being connected to a variety of computer networks with a wireless communication function, a function of transmitting and receiving a variety of data with a wireless communication function, a function of reading a program or data stored in a memory medium and displaying the program or data on the display portion, and the like. Note that functions that can be provided for the electronic devices illustrated in FIGS. 13A to 13G are not limited to those described above, and the electronic devices can have a variety of functions.

Although not shown in FIGS. 13A to 13G; the electronic device may have a plurality of display portions. The electronic device may have a camera or the like and a function of taking a still image, a function of taking a moving image, a function of storing the taken image in a memory medium (an external memory medium or a memory medium incorporated in the camera), a function of displaying the taken image on the display portion, or the like.

(179) The electronic devices shown in FIGS. 13A to 13G will be described in detail.

(180) FIG. 13A is a perspective view of a portable information terminal 9100. A display portion 9001 of the portable information terminal 9100 is flexible. Therefore, the display portion 9001 can be incorporated along a bent surface of a bent housing 9000. In addition, the display portion 9001 includes a touch sensor, and operation can be performed by touching the screen with a finger, a stylus, or the like. For example, when an icon displayed on the display portion 9001 is touched, an application can be started.

(181) FIG. 13B is a perspective view of a portable information terminal 9101. The portable information terminal 9101 functions as, for example, one or more of a telephone set, a notebook, and an information browsing system. Specifically, the portable information terminal can be used as a smartphone. Note that the speaker 9003, the connection terminal 9006, the sensor 9007, and the like, which are not shown in FIG. 13B, can be positioned in the portable information terminal 9101 as in the portable information terminal 9100 shown in FIG. 13A. The portable information terminal 9101 can display characters and image information on its plurality of surfaces. For example, three operation buttons 9050 (also referred to as operation icons, or simply, icons) can be displayed on one surface of the display portion 9001. Furthermore, information 9051 indicated by dashed rectangles can be displayed on another surface of the display portion 9001. Examples of the information 9051 include display indicating reception of an incoming email, social networking service (SNS) message, call, and the like; the title and sender of an email and SNS message; the date; the time; remaining battery; and the reception strength of an antenna. Instead of the information 9051, the operation buttons 9050 or the like may be displayed on the position where the information 9051 is displayed.

(182) FIG. 13C is a perspective view of a portable information terminal 9102. The portable information terminal 9102 has a function of displaying information on three or more surfaces of the display portion 9001. Here, information 9052, 9053, and 9054 are displayed on different surfaces. For example, a user of the portable information terminal 9102 can see the display (here, the information 9053) with the portable information terminal 9102 put in a breast pocket of his/her clothes. Specifically, a caller's phone number, name, or the like of an incoming call is displayed in a position that can be seen from above the portable information terminal 9102. Thus, the user can see the display without taking out the portable information terminal 9102 from the pocket and decide whether to answer the call.

(183) FIG. 13D is a perspective view of a watch-type portable information terminal 9200. The portable information terminal 9200 is capable of executing a variety of applications such as mobile phone calls, e-mailing, viewing and editing texts, music reproduction, Internet communication, and computer games. The display surface of the display portion 9001 is bent, and images can be displayed on the bent display surface. The portable information terminal 9200 can employ near

field communication that is a communication method based on an existing communication standard. In that case, for example, mutual communication between the portable information terminal **9200** and a headset capable of wireless communication can be performed, and thus hands-free calling is possible. The portable information terminal **9200** includes the connection terminal **9006**, and data can be directly transmitted to and received from another information terminal via a connector. Power charging through the connection terminal **9006** is possible. Note that the charging operation may be performed by wireless power feeding without using the connection terminal **9006**.

(184) FIGS. **13E**, **13F**, and **13G** are perspective views of a foldable portable information terminal **9201** that is opened, that is shifted from opened to folded or from folded to opened, and that is folded, respectively. The folded portable information terminal **9201** is highly portable, and the opened portable information terminal **9201** is highly browsable due to a seamless large display region. The display portion **9001** of the portable information terminal **9201** is supported by three housings joined together by hinges **9055**. By folding the portable information terminal **9201** at a connection portion between two housings **9000** with the hinges **9055**, the portable information terminal **9201** can be reversibly changed in shape from opened to folded. For example, the portable information terminal **9201** can be bent with a radius of curvature of greater than or equal to 1 mm and less than or equal to 150 mm.

(185) Electronic devices described in this embodiment are characterized by having a display portion for displaying some sort of information. Note that the light-emitting device of one embodiment of the present invention can also be used for an electronic device which does not have a display portion. The display portion of the electronic device of this embodiment may be non-flexible and display on a flat surface without limitation to the flexible mode capable of displaying along the curved surface or the foldable mode.

(186) The structure described in this embodiment can be used in appropriate combination with the structure described in any of the other embodiments.

Embodiment 7

(187) In this embodiment, examples of lighting devices each using the light-emitting device of one embodiment of the present invention are described with reference to FIG. **14**.

(188) FIG. **14** illustrates an example in which the light-emitting device is used for an interior lighting device **8501**. Note that since the area of the light-emitting device can be increased, a lighting device having a large area can also be formed. In addition, a lighting device **8502** in which a light-emitting region has a curved surface can also be obtained with the use of a housing with a curved surface. A light-emitting element included in the light-emitting device described in this embodiment is in a thin film form, which allows the housing to be designed more freely. Therefore, the lighting device can be elaborately designed in a variety of ways. Further, a wall of the room may be provided with a large-sized lighting device **8503**. Touch sensors may be provided in the lighting devices **8501**, **8502**, and **8503** to control the power on/off of the lighting devices.

(189) Moreover, when the light-emitting device is used at a surface of a table, a lighting device **8504** which has a function as a table can be obtained. When the light-emitting device is used as part of other furniture, a lighting device which has a function as the furniture can be obtained.

(190) In this manner, a variety of lighting devices to which the light-emitting device is applied can be obtained. Note that such lighting devices are also embodiments of the present invention.

(191) Note that the structure described in this embodiment can be combined as appropriate with any of the structures described in the other embodiments.

Example 1

(192) In this example, an example of fabricating a light-emitting element of one embodiment of the present invention will be described. FIGS. **15A** and **15B** are a schematic cross-sectional views of light-emitting elements (light-emitting elements **1** to **4**) fabricated in this example, Table 1 shows the detailed structures of the elements, and structures and abbreviations of compounds used here

are given below.

(193) ##STR00001## ##STR00002##

(194) TABLE-US-00001 TABLE 1 Structure of light-emitting elements of Example 1 Reference Thickness Layer numeral (nm) Material Weight ratio Light- Second electrode 502 200 Al — emitting Electron-injection 516 1 LiF — element 1 layer Electron-transport 515(2) 15 Bphen — layer 515(1) 10 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.8:0.2:0.05 emitting layer PCBBiF: Ir(ppm-dmp).sub.2(acac) First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 40 DBT3P-II:MoOx 2:1 layer First electrode 501 110 ITSO — Light- Second electrode 502 200 Al — emitting Electron-injection 516 1 LiF — element 2 layer Electron-transport 515(2) 15 Bphen — layer 515(1) 10 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.8:0.2:0.05 emitting layer PCBBiF: Ir(ppm-dmp).sub.2(acac) Separation layer 520 2 2mDBTBPDBq-II: 0.6:0.4 PCBBiF First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 40 DBT3P-II:MoOx 2:1 layer First electrode 501 110 ITSO — Light- Second electrode 502 200 Al — emitting Electron-injection 516 1 LiF — element 3 layer Electron-transport 515(2) 15 Bphen — layer 515(1) 10 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.5:0.5:0.05 emitting layer PCBBiF: Ir(ppm-dmp).sub.2(acac) First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 40 DBT3P-II:MoOx 2:1 layer First electrode 501 110 ITSO — Light- Second electrode 502 200 Al — emitting Electron-injection 516 1 LiF — element 4 layer Electron-transport 515(2) 15 Bphen — layer 515(1) 10 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.5:0.5:0.05 emitting layer PCBBiF: Ir(ppm-dmp).sub.2(acac) Separation layer 520 2 2mDBTBPDBq-II: 0.6:0.4 PCBBiF First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 40 DBT3P-II:MoOx 2:1 layer First electrode 501 110 ITSO —

1-1. Fabrication of Light-Emitting Element 1

(195) Indium tin oxide containing silicon oxide (indium tin oxide doped with SiO.sub.2: ITSO) which was formed over a glass substrate **500** to have a thickness of 110 nm and an area of 2 mm×2 mm was used as a first electrode **501**. On the first electrode **501**, 1,3,5-tri(dibenzothiophen-4-yl)benzene (DBT3P-II) and molybdenum oxide (MoO.sub.3) were deposited by co-evaporation in a weight ratio of DBT3P-II:MoO.sub.3=2:1 to a thickness of 40 nm, so that a hole-injection layer **511** was formed. Note that co-evaporation is an evaporation method in which a plurality of different substances is concurrently vaporized from different evaporation sources.

(196) On the hole-injection layer **511**, 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (PCPPn) was deposited by evaporation to a thickness of 20 nm, so that a hole-transport layer **512** was formed.

(197) On the hole-transport layer **512**, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (cgDBCzPA), and N,N'-bis(3-methylphenyl)-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (1,6mMemFLPAPrn) were deposited by co-evaporation in a weight ratio of cgDBCzPA:1,6mMemFLPAPrn=1:0.02 to a thickness of 10 nm, so that the first light-emitting layer **513** was formed. Note that cgDBCzPA was the host material and 1,6mMemFLPAPrn was the fluorescent material (the guest material) in the first light-emitting layer **513**.

(198) On the first light-emitting layer **513**, 2-[3'-(dibenzothiophen-4-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (2mDBTBPDBq-II), N-(1,1'-biphenyl-4-yl)-9,9-dimethyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9H-fluoren-2-amine (PCBBiF), and bis{2-[6-(2,6-dimethylphenyl)-4-pyrimidinyl-κN3]phenyl-κC}(2,4-pentanedionato-κO,O')iridium(III) (Ir(ppm-dmp).sub.2(acac)) were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II:PCBBiF:Ir(ppm-dmp).sub.2(acac)=0.8:0.2:0.05 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed. Note that 2mDBTBPDBq-II was the first organic compound (the

host material), PCBBiF was the second organic compound (the assist material), and Ir(ppm-dmp).sub.2(acac) was the phosphorescent material (the guest material) in the second light-emitting layer **514**.

(199) On the second light-emitting layer **514**, 2mDBTBPDBq-II and bathophenanthroline (Bphen) were sequentially deposited by evaporation to a thickness of 10 nm and 15 nm, respectively, so that electron-transport layers **515(1)** and **512(2)** were formed. On the electron-transport layers **515(1)** and **515(2)**, lithium fluoride was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **516**. Furthermore, aluminum was deposited by evaporation to a thickness of 200 nm to form the second electrode **502**.

(200) Next, a sealing glass substrate was fixed to the glass substrate using a sealing member in a glove box containing a nitrogen atmosphere. In this manner, the light-emitting element was sealed. Note that for sealing, the sealing member was applied to surround the light-emitting element, irradiation with 365-nm ultraviolet light at 6 J/cm² was performed, and heat treatment was performed at 80° C. for 1 hour. Through the above steps, the light-emitting element **1** was obtained.

1-2. Fabrication of Light-Emitting Element 2

(201) Similar to the light emitting element **1**, on the first electrode **501**, DBT3P-II and molybdenum oxide (MoO₃) were deposited by co-evaporation in a weight ratio of DBT3P-II:MoO₃=2:1 to a thickness of 40 nm, so that the hole-injection layer **511** was formed.

(202) On the hole-injection layer **511**, PCPPn was deposited by evaporation to a thickness of 20 nm, so that the hole-transport layer **512** was formed.

(203) On the hole-transport layer **512**, cgDBCzPA and 1,6mMemFLPAPrn were deposited by co-evaporation in a weight ratio of cgDBCzPA:1,6mMemFLPAPrn=1:0.02 to a thickness of 10 nm, so that the first light-emitting layer **513** was formed.

(204) On the first light-emitting layer **513**, 2mDBTBPDBq-II and PCBBiF were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II:PCBBiF=0.6:0.4 to a thickness of 2 nm, so that the separation layer **520** was formed.

(205) On the separation layer **520**, 2mDBTBPDBq-II, PCBBiF, and Ir(ppm-dmp).sub.2(acac) were deposited by co-evaporation in weight ratio of 2mDBTBPDBq-II:PCBBiF:Ir(ppm-dmp).sub.2(acac)=0.8:0.2:0.05 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed.

(206) On the second light-emitting layer **514**, 2mDBTBPDBq-II and Bphen were sequentially deposited by evaporation to a thickness of 10 nm and 15 nm, respectively, so that electron-transport layers **515(1)** and **512(2)** were formed. On the electron-transport layers **515(1)** and **515(2)**, lithium fluoride was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **516**. Furthermore, aluminum was deposited by evaporation to a thickness of 200 nm to form the second electrode **502**.

(207) Next, a sealing glass substrate was fixed to the glass substrate using a sealing member in a glove box containing a nitrogen atmosphere to seal the light-emitting element. In this manner, the light-emitting element **2** was obtained. As the sealing method, a method similar to that used for the light-emitting element **1** was used.

1-3. Fabrication of Light-Emitting Element 3

(208) The light-emitting element **3** was fabricated through the same steps as those for the above-mentioned light-emitting element **1** except steps mentioned below.

(209) On the first light-emitting layer **513**, 2-[3'-(dibenzothiophen-4-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (2mD TBPDBq-II), N-(1,1'-biphenyl-4-yl)-9,9-dimethyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9H-fluoren-2-amine (PCBBiF), and bis{2-[6-(2,6-dimethylphenyl)-4-pyrimidinyl-κN3]phenyl-KC}(2,4-pentanedionato-κO,O')iridium(III) Ir(ppm-dmp).sub.2(acac) were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II:PCBBiF:Ir(ppm-dmp).sub.2(acac)=0.5:0.5:0.05 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed.

1-4. Fabrication of Light-Emitting Element 4

(210) The light-emitting element **4** was fabricated through the same steps as those for the above-mentioned light-emitting element **2** except steps mentioned below.

(211) On the separation layer **520**, 2-[3'-(dibenzothiophen-4-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (2mDB TBPDB q-II), N-(1,1'-biphenyl-4-yl)-9,9-dimethyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9H-fluoren-2-amine (PCBBiF), and bis{2-[6-(2,6-dimethylphenyl)-4-pyrimidinyl-κN3]phenyl-κC}(2,4-pentanedionato-κO,O')iridium(III) Ir(ppm-dmp).sub.2(acac) were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II:PCBBiF:Ir(ppm-dmp).sub.2(acac)=0.5:0.5:0.05 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed.

(212) It is to be noted that an evaporation method using resistive heating was employed for all the evaporation steps.

1-5. Characteristics of Light-Emitting Elements 1 to 4

(213) FIG. **16A** shows current density-luminance characteristics of the light-emitting elements **1** to **4**. In FIG. **16A**, the horizontal axis represents current density (mA/cm.sup.2) and the vertical axis represents luminance (cd/m.sup.2). FIG. **16B** shows voltage-luminance characteristics of the light-emitting elements **1** to **4**. In FIG. **16B**, the horizontal axis represents voltage (V) and the vertical axis represents luminance (cd/m.sup.2). FIG. **17A** shows luminance power efficiency characteristics of the light-emitting elements **1** to **4**. In FIG. **17A**, the horizontal axis represents luminance (cd/m.sup.2) and the vertical axis represents power efficiency (lm/W). FIG. **17B** shows luminance-current efficiency characteristics of the light-emitting elements **1** to **4**. In FIG. **17B**, the horizontal axis represents luminance (cd/m.sup.2) and the vertical axis represents current efficiency (cd/A). Note that the measurement for each light-emitting element was carried out at room temperature (in the atmosphere maintained at 25° C.).

(214) Table 2 shows element characteristics of the light-emitting elements **1** to **4** at around 1000 cd/m.sup.2.

(215) TABLE-US-00002 TABLE 2 Element characteristics of light-emitting elements of Example 1

	External Current efficiency	Current efficiency	Power efficiency	quantum efficiency	Voltage (V)	Current density (mA)	Chromaticity (x, y)	Luminance (cd/m.sup.2)	Current density (mA/cm.sup.2)	Chromaticity (x, y)	Luminance (cd/m.sup.2)	Current density (mA/cm.sup.2)	Chromaticity (x, y)	Luminance (cd/m.sup.2)
Light-emitting element 1	3.1	0.15	3.8	(0.30, 0.34)	910	24	24	10	1000	3.1	0.14	3.4	(0.32, 0.37)	960
Light-emitting element 2	3.1	0.093	2.3	(0.37, 0.44)	870	37	38	12	1000	3.2	0.12	3.0	(0.38, 0.45)	1200
Light-emitting element 3	3.1	0.093	2.3	(0.37, 0.44)	870	37	38	12	1000	3.2	0.12	3.0	(0.38, 0.45)	1200
Light-emitting element 4	3.2	0.12	3.0	(0.38, 0.45)	1200	40	39	13	1000	3.2	0.12	3.0	(0.38, 0.45)	1200

(216) FIG. **18** shows emission spectra when a current at a current density of 2.5 mA/cm.sup.2 was supplied to the light-emitting elements **1** to **4**. As shown in FIG. **18**, each spectrum of the light-emitting elements **1** to **4** has peaks at the blue wavelength region and the yellow wavelength region; therefore, it is found that the two light-emitting materials in each light-emitting element emitted light at a time.

(217) As shown in Table 1, difference between the light-emitting elements **1** and **2** was whether with or without the separation layer **520**, and difference between the light-emitting elements **3** and **4** was whether with or without the separation layer **520**. On the basis of the results shown in FIGS. **16A** and **16B** and FIGS. **17A** and **17B**, the light-emitting elements **1** and **2** had approximately the same element characteristics regardless of whether with or without the separation layer **520**. In addition, on the basis of the results shown in FIGS. **16A** and **16B** and FIGS. **17A** and **17B**, the light-emitting elements **3** and **4** had approximately the same element characteristics regardless of whether with or without the separation layer **520**. The concentration of PCBBiF, which was the second organic compound (the assist material) of the second light-emitting layer **514** of the light-emitting elements **3** and **4**, was high in the light-emitting elements **3** and **4** as compared with that in the light-emitting elements **1** and **2**. Thus, the intensity of the yellow spectrum was increased. Therefore, it is found that the intensity of the yellow spectrum can be easily adjusted while the element characteristics having high efficiency are kept. In addition, on the basis of the results

shown in FIG. 18, it is found that, in the light-emitting elements **1** and **2**, the intensity of the blue and yellow emission spectra can be adjusted by the presence or absence of the separation layer **520**. (218) The structures described in this example can be used in an appropriate combination with any of the structures described in the other embodiments and examples.

Example 2

(219) In this example, an example of fabricating a light-emitting element of one embodiment of the present invention will be described. FIG. 15B is a schematic cross-sectional view of a light-emitting element (a light-emitting element **5**) fabricated in this example, and Table 3 shows the detailed structure of the element, and structures and names of compounds used here are given below. Note that the structures and names of the compounds used for the light-emitting elements **1** to **4** described in Example 1 are not given here.

(220) ##STR00003##

(221) TABLE-US-00003 TABLE 3 Structure of light-emitting element of Example 2 Reference Thickness Layer numeral (nm) Material Weight ratio Light- Second electrode 502 200 Al — emitting Electron-injection 516 1 LiF — element 5 layer Electron-transport 515(2) 15 Bphen — layer 515(1) 10 2mDBTBPDq-II — Second light- 514 20 2mDBTBPDq-II: 0.8:0.2:0.05 emitting layer PCBBiF: Ir(dmppm-dmp).sub.2(acac) Separation layer 520 2 2mDBTBPDq-II: 0.6:0.4 PCBBiF First light-emitting 513 5 cgDBCzPA: 1:0.03 layer 1,6FrAPrn-II Hole-transport layer 512 20 PCPPn — Hole-injection layer 511 20 DBT3P-II:MoO_x 2:1 First electrode 501 110 ITSO —

2-1. Fabrication of Light-Emitting Element 5

(222) Similar to the light emitting element **1**, on the first electrode **501**, DBT3P-II and molybdenum oxide (MoO₃) were deposited by co-evaporation in a weight ratio of DBT3P-II:MoO₃=2:1 to a thickness of 20 nm, so that the hole-injection layer **511** was formed.

(223) On the hole-injection layer **511**, PCPPn was deposited by evaporation to a thickness of 20 nm, so that the hole-transport layer **512** was formed.

(224) On the hole-transport layer **512**, cgDBCzPA and N,N'-bis(dibenzofuran-4-yl)-N,N'-diphenylpyrene-1,6-diamine (1,6FrAPrn-II) were deposited by co-evaporation in a weight ratio of cgDBCzPA:1,6FrAPrn-II=1:0.03 to a thickness of 5 nm, so that the first light-emitting layer **513** was formed. Note that cgDBCzPA was the host material and 1,6FrAPrn-II was the fluorescent material (the guest material) in the first light-emitting layer **513**.

(225) On the first light-emitting layer **513**, 2mDBTBPDq-II and PCBBiF were deposited by co-evaporation in a weight ratio of 2mDBTBPDq-II: PCBBiF=0.6:0.4 to a thickness of 2 nm, so that the separation layer **520** was formed.

(226) On the separation layer **520**, 2mDBTBPDq-II, PCBBiF, and bis{4,6-dimethyl-2-[6-(2,6-dimethylphenyl)-4-pyrimidinyl-κN₃]phenyl-κC}(2,4-pentanedionato-κ^{sup}.2O,O')iridium(III) (Ir(dmppm-dmp).sub.2(acac)) were deposited by co-evaporation in a weight ratio of 2mDBTBPDq-II:PCBBiF:Ir(dmppm-dmp).sub.2(acac)=0.8:0.2:0.05 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed. Note that 2mDBTBPDq-II was the first organic compound (the host material), PCBBiF was the second organic compound (the assist material), and Ir(dmppm-dmp).sub.2(acac) was the phosphorescent material (the guest material) in the second light-emitting layer **514**.

(227) On the second light-emitting layer **514**, 2mDBTBPDq-II and Bphen were sequentially deposited by evaporation to a thickness of 10 nm and 15 nm, respectively, so that electron-transport layers **515(1)** and **512(2)** were formed. On the electron-transport layers **515(1)** and **515(2)**, lithium fluoride was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **516**. Furthermore, aluminum was deposited by evaporation to a thickness of 200 nm to form the second electrode **502**.

(228) It is to be noted that an evaporation method using resistive heating was employed for all the evaporation steps.

(229) Next, a sealing glass substrate was fixed to the glass substrate using a sealing member in a glove box containing a nitrogen atmosphere to seal the light-emitting element. In this manner, the light-emitting element 5 was obtained. As the sealing method, a method similar to that used for the light-emitting element 1 was used.

2-2. Characteristics of Light-Emitting Element 5

(230) FIG. 19A shows current density-luminance characteristics of the light-emitting element 5. In FIG. 19A, the horizontal axis represents current density (mA/cm²) and the vertical axis represents luminance (cd/m²). FIG. 19B shows the voltage-luminance characteristics of the light-emitting element 5. In FIG. 19B, the horizontal axis represents voltage (V) and the vertical axis represents luminance (cd/m²). FIG. 20A shows luminance-power efficiency characteristics of the light-emitting element 5. In FIG. 20A, the horizontal axis represents luminance (cd/m²) and the vertical axis represents power efficiency (lm/W). FIG. 20B shows luminance-current efficiency characteristics of the light-emitting element 5. In FIG. 20B, the horizontal axis represents luminance (cd/m²) and the vertical axis represents current efficiency (cd/A). Note that the measurement for the light-emitting element was carried out at room temperature (under an atmosphere in which the temperature was kept at 25° C.).

(231) Further, Table 4 shows the element characteristics of the light-emitting element 5 at around 1000 cd/m².

(232) TABLE-US-00004 TABLE 4 Element characteristics of light-emitting element of Example 2

	External Current efficiency (%)	Current density (mA/cm ²)	Power efficiency (lm/W)	Voltage (V)	Current efficiency (cd/A)	Chromaticity (x, y)	Luminance (cd/m ²)
Light-emitting element 5	2.9	0.062	1.5	(0.47, 0.45)	870	56 61	18

(233) FIG. 21 shows an emission spectrum when a current at a current density of 2.5 mA/cm² was supplied to the light-emitting element 5. As shown in FIG. 21, the spectrum of the light-emitting element 5 has peaks at the blue wavelength region and the yellow wavelength region; therefore, it is found that the two light-emitting materials therein emitted light at a time.

(234) The light-emitting element 5 of this example was different from the light-emitting elements 1 to 4 of Example 1 in the fluorescent material of the first light-emitting layer 513 and the phosphorescent material of the second light-emitting layer 514. On the basis of the results shown in FIGS. 19A and 19B and FIGS. 20A and 20B, it is found that the light-emitting element 5 had element characteristics with high efficiency, similar to the light-emitting elements 1 to 4 of Example 1. The correlated color temperature of the light-emitting element 5 was 2850 K at around 1000 cd/m². Therefore, it can be used also for a lighting purpose.

(235) The structures described in this example can be used in an appropriate combination with any of the structures described in the other embodiments and examples.

Example 3

(236) In this example, an example of fabricating a light-emitting element of one embodiment of the present invention will be described. FIG. 15B is a schematic cross-sectional view of a light-emitting element (a light-emitting element 6) fabricated in this example, and Table 5 shows the detailed structure of the element, and a structure and a name of a compound used here are given below. Note that the structures and names of the compounds used for the light-emitting elements 1 to 4 described in Example 1 are not given below.

(237) ##STR00004##

(238) TABLE-US-00005 TABLE 5 Structure of light-emitting element of Example 3

Reference	Thickness (nm)	Layer numeral	Material	Weight ratio	Light-emitting element
Second electrode	502	200	Al	—	—
Electron-injection layer	516	1	LiF	—	—
element 6 layer	515(2)	15	Bphen	—	—
Electron-transport layer	515(1)	10	2mDBTBPDBq-II	—	—
Second light-emitting layer	514	20	2mDBTBPDBq-II:	0.8:0.2:0.05	—
Separation layer	520	2	2mDBTBPDBq-II:	0.6:0.4	—
PCBBiF First light-emitting layer	513	5	cgDBCzPA:	1:0.03	—
1,6mMemFLPAPrn Hole-transport layer	512	20	PCPPn	—	—
Hole-injection layer	511	20	DBT3P-II:MoOx	2:1	—
First electrode	—	—	—	—	—

3-1. Fabrication of Light-Emitting Element 6

(239) Similar to the light emitting element **1**, on the first electrode **501**, DBT3P-II and molybdenum oxide (MoO.sub.3) were deposited by co-evaporation in a weight ratio of DBT3P-II:MoO.sub.3=2:1 to a thickness of 20 nm, so that the hole-injection layer **511** was formed.

(240) On the hole-injection layer **511**, PCPPn was deposited by evaporation to a thickness of 20 nm, so that the hole-transport layer **512** was formed.

(241) On the hole-transport layer **512**, cgDBCzPA and 1,6mMemFLPAPrn were deposited by co-evaporation in a weight ratio of cgDBCzPA:1,6mMemFLPAPrn=1:0.03 to a thickness of 5 nm, so that the first light-emitting layer **513** was formed.

(242) On the first light-emitting layer **513**, 2mDBTBPDBq-II and PCBBiF were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II: PCBBiF=0.6:0.4 to a thickness of 2 nm, so that the separation layer **520** was formed.

(243) On the separation layer **520**, 2mDBTBPDBq-II, PCBBiF, and (acetylacetonato)bis[5-methyl-6-(2-methylphenyl)-4-phenylpyrimidinato]iridium(III) (another name: bis {2-[5-methyl-6-(2-methylphenyl)-4-pyrimidinyl-κN3]phenyl-κC}(2,4-pentanedionato-κ.sup.2O,O')iridium(III)) (Ir(mppm).sub.2(acac)) were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II:PCBBiF:Ir(mppm).sub.2(acac)=0.8:0.2:0.05 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed. Note that 2mDBTBPDBq-II was the first organic compound (the host material), PCBBiF was the second organic compound (the assist material), and Ir(mppm).sub.2(acac) was the phosphorescent material (the guest material) in the second light-emitting layer **514**.

(244) On the second light-emitting layer **514**, 2mDBTBPDBq-II and Bphen were sequentially deposited by evaporation to a thickness of 10 nm and 15 nm, respectively, so that electron-transport layers **515(1)** and **512(2)** were formed. On the electron-transport layers **515(1)** and **515(2)**, lithium fluoride was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **516**. Furthermore, aluminum was deposited by evaporation to a thickness of 200 nm to form the second electrode **502**.

(245) It is to be noted that an evaporation method using resistive heating was employed for all the evaporation steps.

(246) Next, a sealing glass substrate was fixed to the glass substrate using a sealing member in a glove box containing a nitrogen atmosphere to seal the light-emitting element. In this manner, the light-emitting element **6** was obtained. As the sealing method, a method similar to that used for the light-emitting element **1** was used.

3-2. Characteristics of Light-Emitting Element 6

(247) FIG. 22A shows current density-luminance characteristics of the light-emitting element **6**. In FIG. 22A, the horizontal axis represents current density (mA/cm.sup.2) and the vertical axis represents luminance (cd/m.sup.2). FIG. 22B shows the voltage-luminance characteristics of the light-emitting element **6**. In FIG. 22B, the horizontal axis represents voltage (V) and the vertical axis represents luminance (cd/m.sup.2). FIG. 23A shows luminance-power efficiency characteristics of the light-emitting element **6**. In FIG. 23A, the horizontal axis represents luminance (cd/m.sup.2) and the vertical axis represents power efficiency (lm/W). FIG. 23B shows luminance-current efficiency characteristics of the light-emitting element **6**. In FIG. 23B, the horizontal axis represents luminance (cd/m.sup.2) and the vertical axis represents current efficiency (cd/A). Note that the measurement for the light-emitting element was carried out at room temperature (under an atmosphere in which the temperature was kept at 25° C.).

(248) Further, Table 6 shows the element characteristics of the light-emitting element **6** at around 1000 cd/m.sup.2.

(249) TABLE-US-00006 TABLE 6 Element characteristics of light-emitting element of Example 3

External Current	Current	Power	quantum	Voltage	Current density	Chromaticity	Luminance
------------------	---------	-------	---------	---------	-----------------	--------------	-----------

efficiency efficiency efficiency (V) (mA) (mA/cm.²) (x, y) (cd/m.²) (cd/A) (lm/W) (%)

Light- 2.9 0.055 1.4 (0.46, 0.47) 790 58 62 19 emitting element 6

(250) FIG. 24 shows an emission spectrum when a current at a current density of 2.5 mA/cm.² was supplied to the light-emitting element 6. As shown in FIG. 24, the spectrum of the light-emitting element 6 has peaks at the blue wavelength region and the yellow wavelength region; therefore, it is found that the two light-emitting materials therein emit light at a time.

(251) The light-emitting element 6 of this example is different from the light-emitting elements 1 to 4 of Example 1 in the phosphorescent material of the second light-emitting layer 514. On the basis of the results shown in FIGS. 22A and 22B and FIGS. 23A and 23B, it is found that the light-emitting element 6 has element characteristics with high efficiency, similar to the light-emitting elements 1 to 4 of Example 1. The correlated color temperature of the light-emitting element 6 was 3090 K at around 1000 cd/m.². Therefore, it can be used also for a lighting purpose.

(252) The structures described in this example can be used in an appropriate combination with any of the structures described in the other embodiments and examples.

Example 4

(253) In this example, an example of fabricating a light-emitting element of one embodiment of the present invention will be described. FIG. 25 is a schematic cross-sectional view of light-emitting elements (light-emitting elements 7 to 10) fabricated in this example, and Table 7 shows the detailed structures of the elements. Note that the structures of compounds used for the light-emitting elements 7 to 10 are the same as those of the compounds used for the light-emitting elements 1 to 6; thus the descriptions thereof are omitted.

(254) TABLE-US-00007 TABLE 7 Structure of light-emitting element of Example 4 Reference Thickness Layer numeral (nm) Material Weight ratio Light-emitting — 552 2360 CF(Red) — Element 7 Second electrode 502(2) 70 ITO 0.5:0.05 *.sup.1) 502(1) 15 Ag:Mg Electron-injection 516 1 LiF — layer Electron-transport 515(2) 20 Bphen — layer 515(1) 15 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.8:0.2:0.06 emitting layer PCBBiF:

Ir(mppm).sub.2(acac) Separation layer 520 2 2mDBTBPDBq-II: 0.2:0.3 PCBBiF First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 87.5 DBT3P-II:MoOx 2:1 layer First electrode 501(3) 75 ITSO — 501(2) 6 Ti — 501(1) 200 Al—Ni—La — Light-emitting — 552 1290 CF(Green) — Element 8 Second electrode 502(2) 70 ITO — 502(1) 15 Ag:Mg 0.5:0.05 *.sup.1) Electron-injection 516 1 LiF — layer Electron-transport 515(2) 20 Bphen — layer 515(1) 15 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.8:0.2:0.06 emitting layer PCBBiF: Ir(mppm).sub.2(acac) Separation layer 520 2 2mDBTBPDBq-II: 0.2:0.3 PCBBiF First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 57.5 DBT3P-II:MoOx 2:1 layer First electrode 501(3) 75 ITSO — 501(2) 6 Ti — 501(1) 200 Al—Ni—La — Light-emitting — 552 780 CF(Blue) — Element 9 Second electrode 502(2) 70 ITO 0.5:0.05 *.sup.1) 502(1) 15 Ag:Mg Electron-injection 516 1 LiF — layer Electron-transport 515(2) 20 Bphen — layer 515(1) 15 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.8:0.2:0.06 emitting layer PCBBiF: Ir(mppm).sub.2(acac) Separation layer 520 2 2mDBTBPDBq-II: 0.2:0.3 PCBBiF First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 50 DBT3P-II:MoOx 2:1 layer First electrode 501(3) 40 ITSO — 501(2) 6 Ti — 501(1) 200 Al—Ni—La — Light-emitting — 552 800 CF (Yellow) — Element 10 Second electrode 502(2) 70 ITO — 502(1) 15 Ag:Mg 0.5:0.05 *.sup.1) Electron-injection 516 1 LiF — layer Electron-transport 515(2) 20 Bphen — layer 515(1) 15 2mDBTBPDBq-II — Second light- 514 20 2mDBTBPDBq-II: 0.8:0.2:0.06 emitting layer PCBBiF: Ir(mppm).sub.2(acac) Separation layer 520 2 2mDBTBPDBq-II: 0.2:0.3 PCBBiF First light-emitting 513 10 cgDBCzPA: 1:0.02 layer 1,6mMemFLPAPrn Hole-transport 512 20 PCPPn — layer Hole-injection 511 65 DBT3P-II:MoOx 2:1 layer First electrode 501(3) 75 ITSO — 501(2) 6 Ti — 501(1) 200 Al—Ni—La — *.sup.1) The ratio of Ag:Mg is

described by the volume ratio.

4-1. Fabrication of Light-Emitting Elements **7** to **10**

(255) On the glass substrate **500**, an alloy film (Al—Ni—La) of aluminum (Al), nickel (Ni), and lanthanum (La) was formed by a sputtering method to a thickness of 200 nm as a first electrode **501(1)**. Next, a titanium (Ti) film was formed by a sputtering method to a thickness of 6 nm and heated at 300° C. for 1 hour to form a film including a titanium oxide as a first electrode **501(2)**. Next, as a first electrode **501(3)**, an indium tin oxide film containing silicon oxide (ITSO) was formed by a sputtering method. Note that the first electrodes **501(1)**, **501(2)**, and **501(3)** formed the first electrode **501**, and the electrode area of the first electrode **501** was 2 mm×2 mm.

(256) The film thickness of the first electrode **501(3)** in each of the light emitting elements **7**, **8**, and **10** was 75 nm, while the film thickness of the first electrode **501(3)** in the light emitting element **9** was 40 nm.

(257) Next, on the first electrode **501(3)**, DBT3P-II and molybdenum oxide (MoO₃) were deposited by co-evaporation in a weight ratio of DBT3P-II:MoO₃=2:1, so that the hole-injection layer **511** was formed.

(258) The film thickness of the hole-injection layer **511** in the light-emitting element **7**, that in the light-emitting element **8**, that in the light-emitting element **9**, and that in the light-emitting element **10** were 87.5 nm, 57.5 nm, 50 nm, and 65 nm, respectively.

(259) Next, on the hole-injection layer **511**, PCPPn was deposited by evaporation to a thickness of 20 nm, so that the hole-transport layer **512** was formed.

(260) On the hole-transport layer **512**, cgDBCzPA and 1,6mMemFLPAPrn were deposited by co-evaporation in a weight ratio of cgDBCzPA:1,6mMemFLPAPrn=1:0.02 to a thickness of 10 nm, so that the first light-emitting layer **513** was formed.

(261) On the first light-emitting layer **513**, 2mDBTBPDBq-II and PCBBiF were deposited by co-evaporation in a weight ratio of 2mDBTBPDBq-II:PCBBiF=0.2:0.3 to a thickness of 2 nm, so that the separation layer **520** was formed.

(262) On the separation layer **520**, 2mDBTBPDBq-II, PCBBiF, and Ir(mppm).sub.2(acac) were deposited by co-evaporation in weight ratio of 2mDBTBPDBq-II:PCBBiF:Ir(mppm).sub.2(acac)=0.8:0.2:0.06 to a thickness of 20 nm, so that the second light-emitting layer **514** was formed.

(263) On the second light-emitting layer **514**, 2mDBTBPDBq-II and Bphen were sequentially deposited by evaporation to a thickness of 15 nm and 20 nm, respectively, so that electron-transport layers **515(1)** and **515(2)** were formed. On the electron-transport layers **515(1)** and **515(2)**, lithium fluoride was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **516**.

(264) On the electron-injection layer **516**, an alloy film of silver (Ag) and magnesium (Mg) was deposited by co-evaporation in a volume ratio of Ag:Mg=0.5:0.05 to a thickness of 15 nm, so that the second electrode **502(1)** was formed.

(265) Next, on the second electrode **502(1)**, an ITO film was formed by a sputtering method to a thickness of 70 nm.

(266) As shown in Table 7, as a coloring layer **552** on a sealing substrate **550**, a red (R) color filter with a thickness of 2.36 μm, a green (G) color filter with a thickness of 1.29 μm, a blue (B) color filter with a thickness of 0.78 μm, and a yellow (Y) color filter with a thickness of 0.80 μm were formed in the light-emitting elements **7**, **8**, **9**, and **10**, respectively.

(267) Each of the light-emitting elements **7** to **10** formed as described above and the corresponding sealing substrate formed as described above were attached with each other to be sealed in a glove box containing a nitrogen atmosphere without exposed to an air atmosphere (the sealing member was applied to surround the element, irradiation with 365-nm ultraviolet light at 6 J/cm² was performed, and heat treatment was performed at 80° C. for 1 hour).

(268) It is to be noted that an evaporation method using resistive heating was employed for all the

evaporation steps.

4-2. Characteristics of Light-Emitting Elements 7 to 10

(269) FIG. 26A shows current density-luminance characteristics of the light-emitting elements 7 to 10. In FIG. 26A, the horizontal axis represents current density (mA/cm²) and the vertical axis represents luminance (cd/m²). FIG. 26B shows the voltage-luminance characteristics of the light-emitting elements 7 to 10. In FIG. 26B, the horizontal axis represents voltage (V) and the vertical axis represents luminance (cd/m²). Note that the measurement for each of the light-emitting elements was carried out at room temperature (under an atmosphere in which the temperature was kept at 25° C.).

(270) Further, Table 8 shows the element characteristics of the light-emitting elements 7 to 10 at around 1000 cd/m².

(271) TABLE-US-00008 TABLE 8 Element characteristics of light-emitting elements of Example 4

	Current (mA)	Voltage (V)	Current density (mA/cm ²)	Chromaticity (x, y)	Luminance (cd/m ²)	efficiency (cd/A)
emitting element 7	3.7	0.64	16	(0.66, 0.34)	930	5.8
emitting element 8	3.1	0.12	3.2	(0.33, 0.64)	850	27
emitting element 9	4.8	2.4	61	(0.14, 0.055)	1000	1.6
emitting element 10	3.0	0.078	2.0	(0.41, 0.58)	810	41

(272) FIG. 27 shows emission spectra when a current at a current density of 2.5 mA/cm² was supplied to the light-emitting elements 7 to 10. FIG. 27 shows that the spectrum of the light-emitting element 7 has a peak in the red wavelength region, that the spectrum of the light-emitting element 8 has a peak in the green wavelength region, that the spectrum of the light-emitting element 9 has a peak in the blue wavelength region, and that the spectrum of the light-emitting element 10 has a peak in the yellow wavelength region. Thus, when the light-emitting elements 7 to 10 are used in combination, full-color display can be achieved.

(273) As shown in Table 7, the light-emitting elements 7 to 10 fabricated in this example each included the same first light-emitting layer 513 and the same second light-emitting layer 514. As shown in FIGS. 26A and 26B and FIG. 27, even when the structure of the first light-emitting layer 513 and the second light-emitting layer 514 were common in the light-emitting elements, the light-emitting elements each of which emitted light with a different emission spectrum from the others had high current efficiency and excellent element characteristics.

(274) The structures described in this example can be used in an appropriate combination with any of the structures described in the other embodiments and examples.

Reference Example

(275) A synthesis method of Ir(ppm-dmp).sub.2(acac) used in Examples 1 and 2 will be described. The synthesis scheme is shown below.

(276) ##STR00005##

1. Synthesis of 4-chloro-6-phenylpyrimidine

(277) A mixture of 5.0 g of 4,6-dichloropyrimidine, 4.9 g of phenylboronic acid, 7.1 g of sodium carbonate, 0.34 g of bis(triphenylphosphine)palladium(II)dichloride (Pd (PPh.sub.3).sub.2Cl.sub.2), 20 mL of acetonitrile, and 20 mL of water was heated to reflux by irradiation with microwaves (2.45 GHz, 100 W) under an argon stream for 1 hour. The obtained mixture was subjected to extraction with dichloromethane and purified by silica gel column chromatography (developing solvent: dichloromethane), whereby 1.6 g of 4-chloro-6-phenylpyrimidine was obtained (yield: 23%, a pale yellow solid). Note that the microwave irradiation in this reference example was performed using a microwave synthesis system (Discover, manufactured by CEM Corporation).

2. Synthesis of 4-phenyl-6-(2,6-dimethylphenyl)pyrimidine (Hppm-dmp)

(278) A mixture of 1.6 g of 4-chloro-6-phenylpyrimidine, 1.5 g of 2,6-dimethylphenylboronic acid, 1.8 g of sodium carbonate, 59 mg of Pd (PPh.sub.3).sub.2Cl.sub.2, 20 mL of N,N-dimethylformamide, and 20 mL of water was heated to reflux by irradiation with microwaves (2.45 GHz, 100 W) under an argon stream for 2 hours. The obtained mixture was subjected to extraction

with dichloromethane and purified by silica gel column chromatography (developing solvent: ethyl acetate and hexane in a ratio of 1:5), whereby 0.50 g of Hppm-dmp was obtained (yield: 23%, a pale yellow oily substance).

3. Synthesis of Di- μ -chloro-tetrakis{2-[6-(2,6-dimethylphenyl)-4-pyrimidinyl- κ N3]phenyl- κ C}diiridium (III) ([Ir(ppm-dmp).SUB.2.Cl].SUB.2.)

(279) A mixture of 1.0 g of Hppm-dmp, 0.57 g of iridium(III) chloride hydrate, 20 mL of 2-ethoxyethanol, and 20 mL of water was heated to reflux by irradiation with microwaves (2.45 GHz, 100 W) under an argon stream for 3 hours. The obtained mixture was filtrated and the resulting solid was washed with methanol, whereby 1.1 g of [Ir(ppm-dmp).sub.2Cl].sub.2 was obtained (yield: 74%, an orange solid).

4. Synthesis of Ir(ppm-dmp).SUB.2.(acac)

(280) A mixture of 1.1 g of [Ir(ppm-dmp).sub.2Cl].sub.2, 0.77 g of sodium carbonate, 0.23 g of acetylacetone (Hacac), and 30 mL of 2-ethoxyethanol was heated to reflux by irradiation with microwaves (2.45 GHz, 120 W) under an argon stream for 2 hours. The obtained mixture was filtrated, and an insoluble was washed with methanol. The obtained filtrate was concentrated, a residue was purified by silica gel column chromatography (developing solvent: ethyl acetate and hexane in a ratio of 1:5), and the obtained solid was recrystallized from hexane, whereby Ir(ppm-dmp).sub.2(acac) was obtained (yield: 59%, an orange powdered solid). By a train sublimation method, 0.21 g of the obtained orange powdered solid were purified, whereby the objective orange solid was collected in a yield of 48%. The conditions of the purification by sublimation were as follows: the pressure was 2.7 Pa; the flow rate of an argon gas was 5.0 mL/min; and the temperature was 240° C. .sup.1H-NMR (nuclear magnetic resonance) spectrum data of the obtained Ir(ppm-dmp).sub.2(acac) are shown below.

(281) .sup.1H-NMR. δ (CDCl.sub.3): 1.85 (s, 6H), 2.26 (s, 12H), 5.35 (s, 1H), 6.46-6.48 (dd, 2H), 6.83-6.90 (dm, 4H), 7.20-7.22 (d, 4H), 7.29-7.32 (t, 2H), 7.63-7.65 (dd, 2H), 7.72 (ds, 2H), 9.24 (ds, 2H).

Example 5

(282) In this example, described are a fabrication example of a light-emitting element **11** not including the second light-emitting layer but including the first light-emitting layer of the light-emitting element of one embodiment of the present invention, and a formation example of a light-emitting element **12** not including the first light-emitting layer but including the second light-emitting layer of the light-emitting element of one embodiment of the present invention. A structure of NN'-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho [1,2-d] furan)-8-amine](abbreviation: 1,6BnfAPrn-03) which is a compound used in this example is given below. The structures and names of the compounds used for the light-emitting elements in the above-described examples are not given below.

(283) ##STR00006##

5-1. Fabrication of Light-Emitting Elements **11** and **12**

(284) Similar to the light-emitting element **1** described in Example 1, the light-emitting element **11** has a structure in which the hole-injection layer **511**, the hole-transport layer **512**, and the first light-emitting layer **513** are stacked over the first electrode **501**; however, the second light-emitting layer **514** is not formed and the electron-transport layers **515(1)** and **515(2)**, the electron-injection layer **516**, and the second electrode **502** are stacked over the first light-emitting layer **513** in this order. Similar to the light-emitting element **1** described in Example 1, the light-emitting element **12** has a structure in which the hole-injection layer **511** and the hole-transport layer **512** are stacked over the first electrode **501**; however, the first light-emitting layer **513** is not formed and the second light-emitting layer **514**, the electron-transport layers **515(1)** and **515(2)**, the electron-injection layer **516**, and the second electrode **502** are stacked over the hole-transport layer **512** in this order.

(285) Thus, Example 1 is referred to for the specific fabrication method of the light-emitting elements. Table 9 shows the specific element structures of the light-emitting elements fabricated in

this example (the light-emitting elements **11** and **12**).

(286) TABLE-US-00009 TABLE 9 Structure of light-emitting elements of Example 5 Reference Thickness Layer numeral (nm) Material Weight ratio Light- Second electrode 502 200 Al — emitting Electron-injection 516 1 LiF — Element layer 11 Electron-transport 515(2) 15 Bphen — layer 515(1) 10 cgDBCzPA — First light-emitting 513 25 cgDBCzPA: 1:0.03 layer 1,6BnfAPrn-03 Hole-transport layer 512 30 PCPPn — Hole injection layer 511 10 PCPPn:MoOx 4:2 First electrode 501 110 ITSO — Light- Second 502 130 Al — emitting electrode Element Electron-injection 516 1 LiF — 12 layer Electron-transport 515(2) 20 Bphen — layer 515(1) 15 2mDBTBPDqII — Second light-emitting 514 40 2mDBTBPDqII: 0.8:0.2:0.06 layer PCBBiF:Ir(mppm).sub.2(acac) Hole-transport layer 512 20 BPAFLP — Hole-injection layer 511 10 DBT3PII:MoOx 1:0.5 First electrode 501 110 ITSO —

5-2. Characteristics of Light-Emitting Elements **11** and **12**

(287) FIG. **28** shows luminance-external quantum efficiency characteristics of the light-emitting element **11**. FIG. **31** shows luminance-external quantum efficiency characteristics of the light-emitting element **12**. In each of FIG. **28** and FIG. **31**, the horizontal axis represents luminance (cd/m.^{sup.2}) and the vertical axis represents external quantum efficiency (%). Note that the measurement of each of the light-emitting elements was carried out at room temperature (under an atmosphere in which the temperature was kept at 25° C.).

(288) Further, Table 10 shows the element characteristics of the light-emitting elements **11** and **12** at around 1000 cd/m.^{sup.2}.

(289) TABLE-US-00010 TABLE 10 Element characteristics of light-emitting elements of Example 5 Current Current External Voltage density Chromaticity Luminance efficiency quantum (V) (mA/cm.^{sup.2}) (x, y) (cd/m.^{sup.2}) (cd/A) efficiency (%) Light- 3.1 9.5 (0.14, 0.13) 1000 11 11 emitting element 11 Light- 2.8 0.88 (0.49, 0.50) 1000 120 32 emitting element 12

(290) FIG. **29** shows an emission spectrum when a current at a current density of 2.5 mA/cm.^{sup.2} was supplied to the light-emitting element **11**. FIG. **32** shows an emission spectrum when a current at a current density of 2.5 mA/cm.^{sup.2} was supplied to the light-emitting element **12**. In each of FIG. **29** and FIG. **32**, the horizontal axis represents wavelength (nm) and the vertical axis represents emission intensity (an arbitrary unit). As shown in FIG. **29**, the emission spectrum of the light-emitting element **11** has a peak in the blue wavelength region, which indicates that emission from the light-emitting material included in the first light-emitting layer **513** was obtained. As shown in FIG. **32**, the emission spectrum of the light-emitting element **12** has a peak in the yellow wavelength region, which indicates that emission from the light-emitting material included in the second light-emitting layer **514** was obtained.

(291) The light-emitting elements **11** and **12** were subjected to reliability tests. FIG. **30** shows the test result of the light-emitting element **11**, and FIG. **33** shows the test result of the light-emitting element **12**. In each of FIG. **30** and FIG. **33**, the vertical axis represents normalized luminance (%) on the assumption that an initial luminance is 100%, and the horizontal axis represents driving time (h) of the light-emitting elements. Note that in the reliability test, the light-emitting elements **11** and **12** were each driven under the conditions where the initial luminance was set to 5000 cd/m.^{sup.2} and the current density was constant. As a result, the light-emitting element **11** kept approximately 90% of the initial luminance until 130 hours had elapsed, and the light-emitting element **12** kept approximately 90% of the initial luminance until 1300 hours had elapsed.

(292) The structures described in this example can be used in an appropriate combination with any of the structures described in the other embodiments and examples.

Reference Example

(293) In this reference example, a method for synthesizing N,N-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho[1,2-d]furan)-8-amine] (abbreviation: 1,6BnfAPrn-03), an organic compound used in this example, is described. Note that a structure of 1,6BnfAPrn-03 is shown below.

(294) ##STR00007##

Step 1: Synthesis of 6-iodobenzo[b]naphtho[1,2-d]furan

(295) Into a 500 mL three-neck flask were put 8.5 g (39 mmol) of benzo[b]naphtho[1,2-d]furan, and the air in the flask was replaced with nitrogen. Then, 195 mL of tetrahydrofuran (THF) was added thereto. This solution was cooled to -75°C . Then, 25 mL (40 mmol) of n-butyllithium (a 1.59 mol/L n-hexane solution) was dropped into this solution. After the drop, the resulting solution was stirred at room temperature for 1 hour.

(296) After a predetermined period of time, the resulting solution was cooled to -75°C . Then, a solution in which 10 g (40 mmol) of iodine had been dissolved in 40 mL of THF was dropped into this solution. After the drop, the resulting solution was stirred for 17 hours while the temperature of the solution was returned to room temperature. After a predetermined period of time, an aqueous solution of sodium thiosulfate was added to the mixture, and the resulting mixture was stirred for 1 hour. Then, an organic layer of the mixture was washed with water and dried with magnesium sulfate. After the drying, the mixture was gravity-filtered to give a solution. The resulting solution was suction-filtered through Celite (Catalog No. 531-16855 produced by Wako Pure Chemical Industries, Ltd.) and Florisil (Catalog No. 540-00135 produced by Wako Pure Chemical Industries, Ltd.) to give a filtrate. The resulting filtrate was concentrated to give a solid. The resulting solid was recrystallized from toluene to give 6.0 g (18 mmol) of white powder of the target substance in a yield of 45%. A synthetic scheme of Step 1 is shown below.

(297) ##STR00008##

Step 2: Synthesis of 6-phenylbenzo[b]naphtho[1,2-d]furan

(298) Into a 200 mL three-neck flask were put 6.0 g (18 mmol) of 6-iodobenzo[b]naphtho[1,2-d]furan, 2.4 g (19 mmol) of phenylboronic acid, 70 mL of toluene, 20 mL of ethanol, and 22 mL of an aqueous solution of potassium carbonate (2.0 mol/L). The mixture was degassed by being stirred while the pressure was reduced. After the degassing, the air in the flask was replaced with nitrogen, and then 480 mg (0.42 mmol) of tetrakis(triphenylphosphine)palladium(0) was added to the mixture. The resulting mixture was stirred at 90°C . under a nitrogen stream for 12 hours.

(299) After a predetermined time has passed, water was added to the mixture, and the solution was separated into the aqueous layer and an organic layer. An extracted solution which was extracted with toluene from the aqueous layer and the organic layer were combined, and the mixture was washed with water and then dried with magnesium sulfate. The mixture was gravity-filtered to give a filtrate. The resulting filtrate was concentrated to give a solid, and the resulting solid was dissolved in toluene. The resulting solution was suction-filtered through Celite (Catalog No. 531-16855 produced by Wako Pure Chemical Industries, Ltd.), Florisil (Catalog No. 540-00135 produced by Wako Pure Chemical Industries, Ltd.), and alumina to give a filtrate. The resulting filtrate was concentrated to give a solid. The resulting solid was recrystallized from toluene to give a 4.9 g (17 mmol) of a white solid of the target substance in a yield of 93%. A synthetic scheme of Step 2 is shown below.

(300) ##STR00009##

Step 3: Synthesis of 8-iodo-6-phenylbenzo[b]naphtho[1,2-d]furan

(301) Into a 300 mL three-neck flask was put 4.9 g (17 mmol) of 6-phenylbenzo[b]naphtho[1,2-d]furan, and the air in the flask was replaced with nitrogen. Then, 87 mL of tetrahydrofuran (THF) was added thereto. The resulting solution was cooled to -75°C . Then, 11 mL (18 mmol) of n-butyllithium (a 1.59 mol/L n-hexane solution) was dropped into the solution. After the drop, the resulting solution was stirred at room temperature for 1 hour. After a predetermined period of time, the resulting solution was cooled to -75°C . Then, a solution in which 4.6 g (18 mmol) of iodine had been dissolved in 18 mL of THF was dropped into the resulting solution.

(302) The resulting solution was stirred for 17 hours while the temperature of the solution was returned to room temperature. After a predetermined period of time, an aqueous solution of sodium thiosulfate was added to the mixture, and the resulting mixture was stirred for 1 hour. Then, an

organic layer of the mixture was washed with water and dried with magnesium sulfate. The mixture was gravity-filtered to give a filtrate. The resulting filtrate was suction-filtered through Celite (Catalog No. 531-16855 produced by Wako Pure Chemical Industries, Ltd.), Florisil (Catalog No. 540-00135 produced by Wako Pure Chemical Industries, Ltd.), and alumina to give a filtrate. The resulting filtrate was concentrated to give a solid. The resulting solid was recrystallized from toluene to give 3.7 g (8.8 mmol) of a target white solid in a yield of 53%. A synthesis scheme of Step 3 is shown below.

(303) ##STR00010##

Step 4: Synthesis of 1,6BnfAPrn-03

(304) Into a 100 mL three-neck flask were put 0.71 g (2.0 mmol) of 1,6-dibromopyrene, 1.0 g (10.4 mmol) of sodium-tert-butoxide, 10 mL of toluene, 0.36 mL (4.0 mmol) of aniline, and 0.3 mL of tri(tert-butyl)phosphine (a 10 wt % hexane solution), and the air in the flask was replaced with nitrogen. To this mixture was added 50 mg (85 μ mol) of bis(dibenzylideneacetone)palladium(0), and the resulting mixture was stirred at 80° C. for 2 hours.

(305) After a predetermined period of time, to the resulting mixture were added 1.7 g (4.0 mmol) of 8-iodo-6-phenylbenzo Plnaphtho[1,2-d]furan, 180 mg (0.44 mmol) of 2-dicyclohexylphosphino-2',6'-dimethoxybiphenyl (abbreviation: S-Phos), and 50 mg (85 μ mol) of bis(dibenzylideneacetone)palladium(0), and the resulting mixture was stirred at 100° C. for 15 hours. After a predetermined period of time, the resulting mixture was filtered through Celite (Catalog No. 531-16855 produced by Wako Pure Chemical Industries, Ltd.) to give a filtrate. The obtained filtrate was concentrated to give a solid. The resulting solid was washed with ethanol and recrystallized from toluene to give 1.38 g (1.4 mmol) of a yellow solid of the target substance in a yield of 71%.

(306) By a train sublimation method, 1.37 mg (1.4 mmol) of the resulting yellow solid was purified by sublimation. The purification by sublimation was performed by heating the yellow solid at 370° C. at an argon flow rate of 10 mL/min under a pressure of 2.3 Pa. As a result of the purification by sublimation, 0.68 g (0.70 mmol) of the yellow solid was obtained in a collection rate of 50%. A synthesis scheme of Step 4 is shown below.

(307) ##STR00011##

(308) An analysis result by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the yellow solid obtained in Step 4 is described below. The result revealed that 1,6BnfAPrn-03 was obtained.

(309) .sup.1H-NMR (dichloromethane-d₂, 500 MHz): δ =6.88 (t, J=7.7 Hz, 4H), 7.03-7.06 (m, 6H), 7.11 (t, J=7.5 Hz, 2H), 7.13 (d, J=8.0 Hz, 2H), 7.28-7.32 (m, 8H), 7.37 (t, J=8.0 Hz, 2H), 7.59 (t, J=7.2 Hz, 2H), 7.75 (t, J=7.7 Hz, 2H), 7.84 (d, J=9.0 Hz, 2H), 7.88 (d, J=8.0 Hz, 2H), 8.01 (s, 2H), 8.07 (d, J=8.0 Hz, 4H), 8.14 (d, J=9.0 Hz, 2H), 8.21 (d, J=8.0 Hz, 2H), 8.69 (d, J=8.5 Hz, 2H).

REFERENCE NUMERALS

(310) **100**: light-emitting element, **101**: electrode, **102**: electrode, **111**: hole-injection layer, **112**: hole-transport layer, **113**: light-emitting layer, **114**: light-emitting layer, **115**: electron-transport layer, **116**: electron-injection layer, **120**: separation layer, **130**: EL layer, **131**: light-emitting unit, **132**: light-emitting unit, **133**: charge generation layer, **140**: light-emitting element, **150**: light-emitting element, **160**: light-emitting element, **170**: light-emitting element, **500**: glass substrate, **501**: electrode, **502**: electrode, **511**: hole-injection layer, **512**: hole-transport layer, **513**: light-emitting layer, **514**: light-emitting layer, **515(1)**: electron-transport layer, **515(2)**: electron-transport layer, **516**: electron-injection layer, **520**: separation layer, **550**: sealing substrate, **552**: coloring layer, **600**: light-emitting device, **601**: source line driver circuit portion, **602**: pixel portion, **603**: gate line driver circuit portion, **604**: sealing substrate, **605**: sealing member, **607**: region, **608**: wiring, **609**: FPC, **610**: element substrate, **611**: FET, **612**: FET, **613**: electrode, **614**: insulator, **616**: EL layer, **617**: electrode, **618**: light-emitting element, **623**: FET, **624**: FET, **801**: pixel circuit, **802**: pixel portion, **804**: driver circuit portion, **804a**: gate driver, **804b**: source driver, **806**: protection circuit, **807**: terminal portion, **852**: transistor, **854**: transistor, **862**: capacitor, **872**: light-emitting

element, **1001**: substrate, **1002**: base insulating film, **1003**: gate insulating film, **1006**: gate electrode, **1007**: gate electrode, **1008**: gate electrode, **1020**: interlayer insulating film, **1021**: interlayer insulating film, **1022**: electrode, **1024B**: electrode, **1024G**: electrode, **1024R**: electrode, **1024Y**: electrode, **1025**: partition, **1026**: electrode, **1028**: EL layer, **1031**: sealing substrate, **1032**: sealing member, **1033**: base material, **1034B**: coloring layer, **1034G**: coloring layer, **1034R**: coloring layer, **1034Y**: coloring layer, **1035**: black layer, **1036**: overcoat layer, **1037**: interlayer insulating film, **1040**: pixel portion, **1041**: driver circuit portion, **1042**: peripheral portion, **8000**: display module, **8001**: upper cover, **8002**: lower cover, **8003**: FPC, **8004**: touch panel, **8005**: FPC, **8006**: display panel, **8009**: frame, **8010**: printed board, **8011**: battery, **8501**: lighting device, **8502**: lighting device, **8503**: lighting device, **8504**: lighting device, **9000**: housing, **9001**: display portion, **9003**: speaker, **9005**: operation key, **9006**: connection terminal, **9007**: sensor, **9008**: microphone, **9050**: operation button, **9051**: information, **9052**: information, **9053**: information, **9054**: information, **9055**: hinge, **9100**: portable information terminal, **9101**: portable information terminal, **9102**: portable information terminal, **9200**: portable information terminal, **9201**: portable information terminal.

(311) This application is based on Japanese Patent Application serial no. 2014-112448 filed with Japan Patent Office on May 30, 2014 and Japanese Patent Application serial no. 2014-241137 filed with Japan Patent Office on Nov. 28, 2014, the entire contents of which are hereby incorporated by reference.

Claims

1. A light-emitting device comprising a pixel portion, the pixel portion comprising: a first transistor comprising a first gate electrode over a first gate insulating film; a second transistor comprising a second gate electrode over a second gate insulating film; a third transistor comprising a third gate electrode over a third gate insulating film; a fourth transistor comprising a fourth gate electrode over a fourth gate insulating film; an insulating film over and in contact with the first gate electrode, the second gate electrode, the third gate electrode, and the fourth gate electrode; a first light-emitting element electrically connected to the first transistor, the first light-emitting element comprising a first anode, a first light-emitting layer over the first anode, a second light-emitting layer over the first light-emitting layer, and a cathode over the second light-emitting layer; a second light-emitting element electrically connected to the second transistor, the second light-emitting element comprising a second anode, the first light-emitting layer over the second anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a third light-emitting element electrically connected to the third transistor, the third light-emitting element comprising a third anode, the first light-emitting layer over the third anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a fourth light-emitting element electrically connected to the fourth transistor, the fourth light-emitting element comprising a fourth anode, the first light-emitting layer over the fourth anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a red color filter under the first anode; a green color filter under the second anode; and a blue color filter under the third anode, wherein a color filter is not provided under the fourth anode, wherein the red color filter, the green color filter, and the blue color filter are in contact with the insulating film, wherein each of the first transistor, the second transistor, the third transistor, and the fourth transistor comprises an oxide semiconductor, wherein each of the first light-emitting element, the second light-emitting element, the third light-emitting element, and the fourth light-emitting element does not comprise a light-emitting layer between the first light-emitting layer and each of the first anode, the second anode, the third anode, and the fourth anode, wherein the first light-emitting layer is separated from the second light-emitting layer, wherein the first light-emitting layer comprises a fluorescent material and a host material,

wherein the second light-emitting layer comprises a phosphorescent material, a first organic compound, and a second organic compound, wherein an emission spectrum of the fluorescent material has a peak in a wavelength region of greater than or equal to 420 nm and less than or equal to 480 nm, wherein an emission spectrum of the phosphorescent material comprises spectral components in a yellow wavelength region and a green wavelength region, wherein a level of the lowest triplet excited state of the host material is lower than a level of the lowest triplet excited state of the first organic compound, wherein the first organic compound and the second organic compound are configured to form an exciplex, and wherein a peak of an emission spectrum of the exciplex overlaps with an absorption band on the longest wavelength side of the phosphorescent material.

2. The light-emitting device according to claim 1, wherein the oxide semiconductor comprises indium.

3. The light-emitting device according to claim 1, wherein the oxide semiconductor comprises indium and zinc.

4. The light-emitting device according to claim 1, wherein the oxide semiconductor comprises indium, gallium, and zinc.

5. The light-emitting device according to claim 1, wherein a light-emitting region of the light-emitting device comprises a curved surface.

6. A light-emitting device comprising a pixel portion, the pixel portion comprising: a first transistor comprising a first gate electrode over a first gate insulating film; a second transistor comprising a second gate electrode over a second gate insulating film; a third transistor comprising a third gate electrode over a third gate insulating film; a fourth transistor comprising a fourth gate electrode over a fourth gate insulating film; an insulating film over and in contact with the first gate electrode, the second gate electrode, the third gate electrode, and the fourth gate electrode; a first light-emitting element electrically connected to the first transistor, the first light-emitting element comprising a first anode, a first light-emitting layer over the first anode, a second light-emitting layer over the first light-emitting layer, and a cathode over the second light-emitting layer; a second light-emitting element electrically connected to the second transistor, the second light-emitting element comprising a second anode, the first light-emitting layer over the second anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a third light-emitting element electrically connected to the third transistor, the third light-emitting element comprising a third anode, the first light-emitting layer over the third anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a fourth light-emitting element electrically connected to the fourth transistor, the fourth light-emitting element comprising a fourth anode, the first light-emitting layer over the fourth anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a red color filter under the first anode; a green color filter under the second anode; and a blue color filter under the third anode, wherein a color filter is not provided under the fourth anode, wherein the red color filter, the green color filter, and the blue color filter are in contact with the insulating film, wherein each of the first transistor, the second transistor, the third transistor, and the fourth transistor comprises an oxide semiconductor, wherein each of the first light-emitting element, the second light-emitting element, the third light-emitting element, and the fourth light-emitting element does not comprise a light-emitting layer between the first light-emitting layer and each of the first anode, the second anode, the third anode, and the fourth anode, wherein the first light-emitting layer is separated from the second light-emitting layer, wherein the first light-emitting layer comprises a fluorescent material and a host material, wherein the second light-emitting layer comprises a phosphorescent material, a first organic compound, and a second organic compound, wherein an emission spectrum of the fluorescent material has a peak in a wavelength region of greater than or equal to 420 nm and less than or equal to 480 nm, wherein an emission spectrum of the phosphorescent material comprises spectral

components in a yellow wavelength region and a wavelength region of greater than or equal to 500 nm and less than 550 nm, wherein a level of the lowest triplet excited state of the host material is lower than a level of the lowest triplet excited state of the first organic compound, wherein the first organic compound and the second organic compound are configured to form an exciplex, and wherein a peak of an emission spectrum of the exciplex overlaps with an absorption band on the longest wavelength side of the phosphorescent material.

7. The light-emitting device according to claim 6, wherein the oxide semiconductor comprises indium.

8. The light-emitting device according to claim 6, wherein the oxide semiconductor comprises indium and zinc.

9. The light-emitting device according to claim 6, wherein the oxide semiconductor comprises indium, gallium, and zinc.

10. The light-emitting device according to claim 6, wherein a light-emitting region of the light-emitting device comprises a curved surface.

11. A light-emitting device comprising a pixel portion, the pixel portion comprising: a first transistor comprising a first gate electrode over a first gate insulating film; a second transistor comprising a second gate electrode over a second gate insulating film; a third transistor comprising a third gate electrode over a third gate insulating film; a fourth transistor comprising a fourth gate electrode over a fourth gate insulating film; an insulating film over and in contact with the first gate electrode, the second gate electrode, the third gate electrode, and the fourth gate electrode; a first light-emitting element electrically connected to the first transistor, the first light-emitting element comprising a first anode, a first light-emitting layer over the first anode, a second light-emitting layer over the first light-emitting layer, and a cathode over the second light-emitting layer; a second light-emitting element electrically connected to the second transistor, the second light-emitting element comprising a second anode, the first light-emitting layer over the second anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a third light-emitting element electrically connected to the third transistor, the third light-emitting element comprising a third anode, the first light-emitting layer over the third anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a fourth light-emitting element electrically connected to the fourth transistor, the fourth light-emitting element comprising a fourth anode, the first light-emitting layer over the fourth anode, the second light-emitting layer over the first light-emitting layer, and the cathode over the second light-emitting layer; a red color filter under the first anode; a green color filter under the second anode; and a blue color filter under the third anode, wherein a color filter is not provided under the fourth anode, wherein the red color filter, the green color filter, and the blue color filter are in contact with the insulating film, wherein each of the first transistor, the second transistor, the third transistor, and the fourth transistor comprises an oxide semiconductor, wherein each of the first light-emitting element, the second light-emitting element, the third light-emitting element, and the fourth light-emitting element does not comprise a light-emitting layer between the first light-emitting layer and each of the first anode, the second anode, the third anode, and the fourth anode, wherein the first light-emitting layer is separated from the second light-emitting layer, wherein the first light-emitting layer comprises a fluorescent material and a host material, wherein the second light-emitting layer comprises a phosphorescent material, a first organic compound, and a second organic compound, wherein an emission spectrum of the fluorescent material has a peak in a wavelength region of greater than or equal to 420 nm and less than or equal to 480 nm, wherein an emission spectrum of the phosphorescent material comprises spectral components in a green wavelength region, a yellow wavelength region, and a wavelength region of greater than or equal to 600 nm and less than or equal to 740 nm, wherein a level of the lowest triplet excited state of the host material is lower than a level of the lowest triplet excited state of the first organic compound, wherein the first organic compound and the second organic compound are

- configured to form an exciplex, and wherein a peak of an emission spectrum of the exciplex overlaps with an absorption band on the longest wavelength side of the phosphorescent material.
12. The light-emitting device according to claim 11, wherein the oxide semiconductor comprises indium.
13. The light-emitting device according to claim 11, wherein the oxide semiconductor comprises indium and zinc.
14. The light-emitting device according to claim 11, wherein the oxide semiconductor comprises indium, gallium, and zinc.
15. The light-emitting device according to claim 11, wherein a light-emitting region of the light-emitting device comprises a curved surface.
-